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Papers 214 (conclusion), 215, and 216 (opening)

Contents

214. On the Development of the Disturbing Function in the Lunar and Planetary Theories *(continued)*

[PDF p. 351; printed p. 329.] For the lunar theory, to the extent to which it is necessary to carry the development, and to which it is carried in Hansen's *Fundamenta Nova*, and my memoir before referred to, we might simply write,

Disturbing function ÷ Sun's Mass is

		U	$+U'$
	$\frac{a^2}{a'^3}\,C_2(0,0)$	cos	0 0
(1)	$+\frac{a^3}{a'^4}\,C_3(0,0)$		0 0
(2)	$+2\,\frac{a^3}{a'^4}\,C_3(1,-1)$		1 − 1
(3)	$+2\,\frac{a^3}{a'^4}\,C_3(1,1)$		1 1
(4)	$+2\,\frac{a^3}{a'^4}\,C_3(2,-2)$		2 − 2
(5)	$+2\,\frac{a^3}{a'^4}\,C_3(2,0)$		2 0
(6)	$+2\,\frac{a^3}{a'^4}\,C_3(0,2)$		0 2
(7)	$+2\,\frac{a^3}{a'^4}\,C_3(2,2)$		2 2
(8)	$+2\,\frac{a^3}{a'^4}\,C_3(3,-3)$		3 − 3
(9)	$+2\,\frac{a^3}{a'^4}\,C_3(3,-1)$		3 − 1
(10)	$+2\,\frac{a^3}{a'^4}\,C_3(1,-3)$		1 − 3
	$+2\,\frac{a^3}{a'^4}\,C_3(3,1)$		3 1
	$+2\,\frac{a^3}{a'^4}\,C_3(1,3)$		1 3

where the prefixed numbers are those of Hansen's two parts Ω_1, Ω_2,... Ω_6; or omitting the first term which depends only on the moon's radius vector, and the last two terms which we ultimately neglected, and substituting for the coefficients C their values we here, [PDF p. 352; printed p. 330.] Disturbing function ÷ Sun's Mass is

		U	$+U'$
(1)	$\dfrac{a}{a'} - \frac{3}{4}\eta^2 \dfrac{a^3}{a'^4} + \frac{3}{8}\eta^4 \dfrac{a^3}{a'^4}$	cos	$0 \quad 0$
(2)	$+\dfrac{a^3}{a'^4}\,\frac{1}{4}\eta^2$		$1 \quad -1$
(3)	$+\dfrac{a^3}{a'^4}\,\frac{1}{4}\eta^2$		$1 \quad 1$
(4)	$+\dfrac{a^3}{a'^4}\,\frac{15}{8}\eta^4$		$2 \quad -2$
(5)	$-\frac{3}{4}\eta^2 \dfrac{a^3}{a'^4} + \frac{15}{8}\eta^4 \dfrac{a^3}{a'^4}$		$2 \quad 0$
(6)	$-\frac{3}{4}\eta^2 \dfrac{a^3}{a'^4} + \frac{15}{8}\eta^4 \dfrac{a^3}{a'^4}$		$0 \quad 2$
(7)	$+\frac{15}{8}\eta^4 \dfrac{a^3}{a'^4}$		$2 \quad 2$
(8)	$+\frac{15}{8}\eta^4 \dfrac{a^3}{a'^4}$		$3 \quad -3$
(9)	$+\frac{15}{8}\eta^4 \dfrac{a^3}{a'^4}$		$3 \quad -1$
(10)	$+\frac{15}{8}\eta^4 \dfrac{a^3}{a'^4}$		$1 \quad -3$

If the same form be adopted for the planetary theory, the expressions for the leading coefficients will be,

$$
\begin{array}{lll}
C_2(0,0), \quad D(0,0) = \frac{1}{2}, & & \\
\qquad\qquad 2 & + \dfrac{a^2}{a'^2}\left(1 - 6\,\eta^2 + 6\,\eta^4\right), & \\
\qquad\qquad 4 & + \dfrac{a^4}{a'^4}\,\frac{9}{32}\left(1 - 20\,\eta^2 + 90\,\eta^4 - 140\,\eta^6 + 70\,\eta^8\right), & \\
\qquad\qquad 6 & + \dfrac{a^6}{a'^6}\,\frac{25}{12\cdot 32}\left(1 - 42\,\eta^2 + 420\,\eta^4 - 1680\,\eta^6 + 3150\,\eta^8 \ldots\right), & \\
\qquad\qquad 8 & + \dfrac{a^8}{a'^8}\,\frac{1225}{2\cdot 16384}\left(1 - 72\,\eta^2 + 1260\,\eta^4 - 9240\,\eta^6 + 34650\,\eta^8 \ldots\right), & \\
\qquad\qquad 2\rho & + \dfrac{a^{2\rho}}{a'^{2\rho}}\,\dfrac{\Pi_1(\rho-1)\,\Pi_1(\rho-\frac{1}{2})}{\Pi_1\rho\,\Pi\rho}\,F\left(-2\rho,\ 2\rho+1,\ 1,\ \eta^2\right); &
\end{array}
$$

[PDF p. 353; printed p. 331.]

$$
\begin{array}{lll}
C_1(1,-1), \quad D(1,-1) = & & \dfrac{a}{a'}\,\tfrac{1}{2}(1-\eta^2) \\
& 3 & +\dfrac{a^3}{a'^3}\,\tfrac{3}{16}(1-10\,\eta^2+\ldots) \\
& 5 & +\dfrac{a^5}{a'^5}\,\tfrac{15}{128}(1-28\,\eta^2+168\,\eta^4-336\,\eta^6+210\,\eta^8\ldots) \\
& 7 & +\dfrac{a^7}{a'^7}\,\tfrac{105}{1024}(1-54\,\eta^2+675\,\eta^4-3100\,\eta^6+7425\,\eta^8\ldots) \\
& 2\rho+1 & +\dfrac{a^{2\rho+1}}{a'^{2\rho+1}}\,\dfrac{\Pi_{\frac12}(2\rho+\frac12)\,\Pi_{\frac12}(\rho-\frac12)}{\Pi_{\frac12}\rho\,\Pi_{\frac12}(\rho+1)}\,(1-\eta^2)\,F(-2\rho,\,2\rho+3,\,3,\,\eta^2);
\end{array}
$$

$$
\begin{array}{lll}
C_1(1,1), \quad D(1,1) = & & \dfrac{a}{a'}\,\tfrac{1}{2}\eta^2 \\
& 3 & +\dfrac{a^3}{a'^3}\,\tfrac{1}{8}\eta^2(1-7\,\eta^2+\tfrac{15}{2}\eta^4) \\
& 5 & +\dfrac{a^5}{a'^5}\,\tfrac{15}{128}\eta^2(1-14\,\eta^2+\tfrac{63}{2}\eta^4-\ldots) \\
& 7 & +\dfrac{a^7}{a'^7}\,\tfrac{105}{1024}\eta^2(1-18\,\eta^2+\ldots) \\
& 2\rho+1 & +\dfrac{a^{2\rho+1}}{a'^{2\rho+1}}\,\dfrac{\Pi_{\frac12}(\rho+\frac12)\,\Pi_{\frac12}(\rho-\frac12)}{\Pi_{\frac12}\rho\,\Pi_{\frac12}(\rho-1)}\,\eta^2\,F(-2\rho,\,2\rho+3,\,3,\,\eta^2);
\end{array}
$$

$$
\begin{array}{lll}
C_2(2,-2), \quad D(2,-2) = & & \dfrac{a^2}{a'^2}\,\tfrac{3}{8}(1-\eta^2)^2 \\
& 4 & +\dfrac{a^4}{a'^4}\,\tfrac{5}{16}(1-\eta^2)^2(1-14\,\eta^2+24\,\eta^4) \\
& 6 & +\dfrac{a^6}{a'^6}\,\tfrac{105}{512}(1-\eta^2)^2(1-36\,\eta^2+360\,\eta^4-660\,\eta^6+495\,\eta^8) \\
& \vdots & \\
& 2\rho & +\dfrac{a^{2\rho}}{a'^{2\rho}}\,\dfrac{\Pi_1(\rho+\frac12)\,\Pi_1(\rho-\frac12)}{\Pi_1(\rho+1)\,\Pi_1(\rho-1)}\,(1-\eta^2)^2\,F(-2\rho+2,\,2\rho+2,\,5,\,\eta^2);
\end{array}
$$

$$
\begin{array}{lll}
C_2(2,0), \quad D(2,0) = & & \dfrac{a^2}{a'^2}\,\tfrac{1}{2}\eta^2(1-\eta^2) \\
& 4 & +\dfrac{a^4}{a'^4}\,\tfrac{5}{8}\eta^2(1-\eta^2)(1-6\,\eta^2+\ldots) \\
& 6 & +\dfrac{a^6}{a'^6}\,\tfrac{105}{32}\eta^2(1-\eta^2)(1-12\,\eta^2+45\,\eta^4-\ldots) \\
& \vdots & \\
& 2\rho & +\dfrac{a^{2\rho}}{a'^{2\rho}}\,\dfrac{\Pi_1(\rho+\frac12)\,\Pi_1(\rho-\frac12)}{\Pi_1\rho\,\Pi_1(\rho-1)}\,\eta^2(1-\eta^2)\,F(-2\rho+2,\,2\rho+2,\,3,\,\eta^2).
\end{array}
$$

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$$C_2(0,2) = C_2(2,0), \qquad D(0,2) = D(2,0);$$

$$\begin{aligned}
C_2(2,2), \quad D(2,2) &= \frac{a^2}{a'^2}\,\tfrac{3}{8}\eta^4 \\
4 \qquad &\quad + \frac{a^4}{a'^4}\,\tfrac{5}{16}\eta^4(1 - 7\,\eta^2 + \tfrac{63}{8}\eta^4) \\
6 \qquad &\quad + \frac{a^6}{a'^6}\,\tfrac{105}{512}\eta^4(1 - 17\,\eta^2 + 18\,\eta^4) \\
\vdots \qquad & \\
2\rho \qquad &\quad + \frac{a^{2\rho}}{a'^{2\rho}}\,\frac{\Pi_1(\rho+\frac{1}{2})\,\Pi_1(\rho-\frac{1}{2})}{\Pi_1(\rho+1)\,\Pi_1(\rho-1)}\,\eta^4\,F(-2\rho+2,\ 2\rho+2,\ 5,\ \eta^2);
\end{aligned}$$

$$\begin{aligned}
C_3(3,-3), \quad D(3,-3) &= \frac{a^3}{a'^3}\,\tfrac{5}{16}(1-\eta^2)^3 \\
5 \qquad &\quad + \frac{a^5}{a'^5}\,\tfrac{35}{128}(1-\eta^2)^3(1 - 18\,\eta^2 + \ldots) \\
\vdots \qquad & \\
2\rho+1 \qquad &\quad + \frac{a^{2\rho+1}}{a'^{2\rho+1}}\,\frac{\Pi_{\frac{1}{2}}(\rho+\frac{3}{2})\,\Pi_{\frac{1}{2}}(\rho-\frac{1}{2})}{\Pi_{\frac{1}{2}}(\rho+2)\,\Pi_{\frac{1}{2}}(\rho-1)}\,(1-\eta^2)^3\,F(-2\rho+2,\ 2\rho+3,\ 7,\ \eta^2);
\end{aligned}$$

$$\begin{aligned}
C_3(3,-1), \quad D(3,-1) &= \frac{a^3}{a'^3}\,\tfrac{15}{8}\eta^2(1-\eta^2)^2 \\
5 \qquad &\quad + \frac{a^5}{a'^5}\,\tfrac{105}{32}\eta^2(1-\eta^2)^2(1 - 6\,\eta^2 + \ldots) \\
\vdots \qquad & \\
2\rho+1 \qquad &\quad + \frac{a^{2\rho+1}}{a'^{2\rho+1}}\,\frac{\Pi_{\frac{1}{2}}(\rho+\frac{3}{2})\,\Pi_{\frac{1}{2}}(\rho-\frac{1}{2})}{\Pi_{\frac{1}{2}}(\rho+1)\,\Pi_{\frac{1}{2}}(\rho-1)}\,\eta^2(1-\eta^2)^2\,F(-2\rho+2,\ 2\rho+3,\ 5,\ \eta^2);
\end{aligned}$$

$$C_3(1,-3) = C_3(3,-1), \qquad D(1,-3) = D(3,-1);$$

$$\begin{aligned}
C_3(3,1), \quad D(3,1) &= \frac{a^3}{a'^3}\,\tfrac{15}{8}\eta^4(1-\eta^2) \\
5 \qquad &\quad + \frac{a^5}{a'^5}\,\tfrac{105}{32}\eta^4(1-\eta^2)(1 - \ldots) \\
\vdots \qquad & \\
2\rho+1 \qquad &\quad + \frac{a^{2\rho+1}}{a'^{2\rho+1}}\,\frac{\Pi_{\frac{1}{2}}(\rho+\frac{3}{2})\,\Pi_{\frac{1}{2}}(\rho-\frac{1}{2})}{\Pi_{\frac{1}{2}}\rho\,\Pi_{\frac{1}{2}}(\rho-1)}\,\eta^4(1-\eta^2)\,F(-2\rho+2,\ 2\rho+3,\ 5,\ \eta^2);
\end{aligned}$$

$$C_3(1,3) = C_3(3,1), \qquad D(1,3) = D(3,1);$$

$$\begin{aligned}
C_3(3,3), \quad D(3,3) &= \frac{a^3}{a'^3}\,\tfrac{5}{16}\eta^6 \\
5 \qquad &\quad + \frac{a^5}{a'^5}\,\tfrac{35}{128}\eta^6(1 + \ldots) \\
\vdots \qquad & \\
2\rho+1 \qquad &\quad + \frac{a^{2\rho+1}}{a'^{2\rho+1}}\,\frac{\Pi_{\frac{1}{2}}(\rho+\frac{3}{2})\,\Pi_{\frac{1}{2}}(\rho-\frac{1}{2})}{\Pi_{\frac{1}{2}}(\rho+2)\,\Pi_{\frac{1}{2}}(\rho-1)}\,\eta^6\,F(-2\rho+2,\ 2\rho+3,\ 7,\ \eta^2) \quad \text{\&c.}
\end{aligned}$$

IV.

[PDF p. 355; printed p. 333.] But for the planetary theory it is more suitable to arrange the expression of $D(j, j')$ according to the powers of η. We have, under the before-mentioned conditions, j not negative and not less in absolute magnitude than j',

$$D(j, j') = \sum_{\lambda=0}^{\infty} \frac{a'^{2\lambda}}{a^{2\lambda}} \, C_{j+2\lambda}(j, j'),$$

where λ extends from 0 to ∞; and, writing in the expression of $C_n(j, j')$, $j + 2\lambda$ for n, we find

$$C_{j+2\lambda}(j, j') = \frac{2^{j+j'} \, \Pi_1\big(\tfrac{1}{2}(j+j') + \lambda - \tfrac{1}{2}\big) \, \Pi_1\big(\tfrac{1}{2}(j-j') + \lambda - \tfrac{1}{2}\big)}{\Pi_1 \, \Pi_1\big(\tfrac{1}{2}(j-j') + \lambda\big) \, \Pi_1(j+j'+\lambda)} \, \eta^{j+j'} \, F(-2\lambda,\ 2j{+}2\lambda{+}1,\ j{+}j'{+}1,\ \eta^2),$$

or, substituting for the hypergeometric series its value,

$$C_{j+2\lambda}(j, j') = \sum_{\theta} \frac{2^{j+j'} \, \Pi_1\big(\tfrac{1}{2}(j+j') + \lambda - \tfrac{1}{2}\big) \, \Pi_1\big(\tfrac{1}{2}(j-j') + \lambda - \tfrac{1}{2}\big)}{\Pi_1 \, \Pi_1\big(\tfrac{1}{2}(j-j') + \lambda\big) \, \Pi_1(j+j'+\lambda)} \, (-)^{\theta} \, \frac{\Pi 2\lambda \, \Pi(2j+2\lambda+\theta)}{\Pi(2\lambda-\theta) \, \Pi\theta \, \Pi(j+j'+\theta) \, \Pi(\ldots}$$

from $\theta = 0$ to $\theta = \infty$. Substituting in the numerator for $\Pi 2\lambda$, $2^{2\lambda} \, \Pi\lambda \, \Pi_1(\lambda - \tfrac{1}{2})$, and in the denominator for $\Pi(2j + 2\lambda)$, $2^{2j+2\lambda} \, \Pi(j+\lambda) \, \Pi(j+\lambda-\tfrac{1}{2})$, and reducing, this becomes

$$C_{j+2\lambda}(j, j') = \sum_{\theta} \frac{(-)^{\theta} \, 2^{-j+j'} \, \Pi_1\big(\tfrac{1}{2}(j+j') + \lambda - \tfrac{1}{2}\big) \, \Pi_1\big(\tfrac{1}{2}(j+j') + \lambda + \theta - \tfrac{1}{2}\big)}{\Pi_1(j+\lambda) \, \Pi_1(j+\lambda-\tfrac{1}{2}) \, \Pi_1\big(\tfrac{1}{2}(j-j') + \lambda\big) \, \Pi(j+j'+\theta) \, \Pi\lambda \, \Pi(2\lambda-\theta)} \, \eta^{j+j'+2\theta},$$

and substituting this expression in $D(j, j')$, we find

$$D(j, j') = \frac{1}{a^{j+j'}} \, \sum \frac{a^{2\lambda}}{a'^{2\lambda}} \, \eta^{j+j'+2\theta} \cdot (\text{the same kernel}),$$

$$\sum \frac{(-)^{\theta} \, 2^{-j+j'} \, \Pi_1\big(\tfrac{1}{2}(j+j') + \lambda - \tfrac{1}{2}\big) \, \Pi_1\big(\tfrac{1}{2}(j+j') + \lambda + \theta - \tfrac{1}{2}\big)}{\Pi_1(j+\lambda) \, \Pi_1(j+\lambda-\tfrac{1}{2}) \, \Pi_1\big(\tfrac{1}{2}(j-j') + \lambda\big) \, \Pi(j+j'+\theta) \, \Pi\lambda \, \Pi(2\lambda-\theta)}$$

from $\lambda = 0$ to $\lambda = \infty$ and $\theta = 0$ to $\theta = \infty$; which is the required expression. It is to be remarked that the series in $\big(\dfrac{a}{a'}\big)$ which multiplies $\eta^{j+j'+2\theta}$ is not in general expressible as a hypergeometric series. If, however, we attend only to the leading term in η, or write $\theta = 0$, we find for $D(j, j')$ the value

$$\frac{1}{a^{j+j'}} \, \eta^{j+j'} \, (1-\eta^2)^{\frac{1}{2}(j-j')} \sum \frac{a^{2\lambda}}{a'^{2\lambda}} \, \frac{\Pi_1\big(\tfrac{1}{2}(j+j') + \lambda - \tfrac{1}{2}\big) \, \Pi(2j+2\lambda)}{\Pi_1(j+\lambda) \, \Pi_1(j+\lambda-\tfrac{1}{2}) \, \Pi_1\big(\tfrac{1}{2}(j-j') + \lambda\big) \, \Pi(j+j') \, \Pi\lambda},$$

[PDF p. 356; printed p. 334.] which may be simplified by putting in the numerator for $\Pi(2j+2\lambda)$,

$$2^{2j+2\lambda}\,\Pi_1(j+\lambda)\,\Pi_1(j+\lambda-\tfrac{1}{2}),$$

and in the denominator for $\Pi_1(j+j')$,

$$\frac{\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda-\tfrac{1}{2}\big)\,\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda\big)}{\Pi_1\big(\tfrac{1}{2}(j+j')-\tfrac{1}{2}\big)\,\Pi_1\big(\tfrac{1}{2}(j+j')\big)},$$

which is equal to

$$\frac{\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda-\tfrac{1}{2}\big)\,\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda\big)}{\Pi_1\big(\tfrac{1}{2}(j-j')-\tfrac{1}{2}\big)\,\Pi_1\big(\tfrac{1}{2}(j+j')\big)};$$

or finally we have, as regards the leading term in η,

$$D(j,j') = \eta^{j+j'}\,(1-\eta^2)^{\frac{1}{2}(j-j')}\,\frac{\Pi_1\big(\tfrac{1}{2}(j+j')\big)}{\Pi_1\,\Pi_1\big(\tfrac{1}{2}(j+j')-\tfrac{1}{2}\big)\,\Pi_1\big(\tfrac{1}{2}(j-j')\big)}\sum\frac{a^{2\lambda}}{a'^{2\lambda}}\,F\big(\tfrac{1}{2}(j+j')+\lambda,\ \tfrac{1}{2}(j+j')+\lambda-\tfrac{1}{2},\ldots$$

I remark that if in general

$$r^{\alpha+\beta}\,r'^{\alpha-\beta}\,(r^2+r'^2-2\,r\,r'\cos\vartheta)^{-\alpha-\frac{1}{2}} = \sum_i R_i^{\alpha,\beta}\cos i\vartheta,$$

then, writing $\frac{1}{2}(j-j')$ for β and $\frac{1}{2}(-j+j')$ for β', we have

$$D(j,j') = R_j^{\alpha,\beta}\,(1-\eta^2)^{\frac{1}{2}(j-j')}\,(\text{terms in }\eta),$$

and the last-mentioned expression for $D(j,j')$, as regards the leading term in η, becomes therefore

$$\eta^{j+j'}\,R_{j+j'}^{\alpha,\beta}\,(1-\eta^2)^{\frac{1}{2}(j-j')},$$

and we have in general

$$D(j,j') = \eta^{j+j'}\,(1-\eta^2)^{\frac{1}{2}(j-j')}\,R_{j+j'}^{\alpha,\beta}\,(\text{terms in }\eta).$$

As a verification, it may be noticed that for $j+j'=0$ we have $D(j,j')=(1-\eta^2)^j\,(R_0+$ terms in $\eta)$, and for $\eta=0$, $D(j,j')=R_j$, which is right, since the two sides each denote the coefficient of $\cos j\vartheta$ in $(r^2+r'^2-2\,r\,r'\cos\vartheta)^{-\frac{1}{2}}$. There is reason to believe that the expression for $D(j,j')$ might be further reduced so as to obtain in a convenient form the coefficients of the successive powers of η; but I have not yet accomplished this.

V.

[PDF p. 357; printed p. 335.] Considering now $D(j,j')$ as a given function of r and r', we must in the planetary theory write $r = a(1+x)$, $r' = a'(1+x')$, and develope in powers of x and x'. The general term is, of course,

$$\frac{a^\alpha\, a'^{\alpha'}}{\Pi a\, \Pi a'}\, \frac{d^{\alpha+\alpha'}\, D(j,j')}{da^\alpha\, da'^{\alpha'}},$$

where $D'(j,j')$ is what $D(j,j')$ becomes when a, a' are substituted for r, r'; and, writing $f+\mathfrak{C}$, $f'+\mathfrak{C}'$ for U, U', we see that in the planetary theory the terms to be developed are of the form

$$\frac{r^n\, r'^{n'}}{a^n\, a'^{n'}} \cos\bigl(jf + j'f' + j\mathfrak{C} + j'\mathfrak{C}'\bigr),$$

while, from what has preceded, the form for the lunar theory is

$$\frac{r^n\, r'^{n'}}{a^n\, a'^{n'}} \cos\bigl(jf + j'f' + j\mathfrak{C} + j'\mathfrak{C}'\bigr),$$

(n is always $= -n' - 1$, except in the terms $\cos(U - U')$); the values which have to be substituted being

$$r = a\ \mathrm{elqr}(e,g), \qquad x = \mathrm{elqr}(e,g) - 1,$$
$$f = \mathrm{elta}(e,g),$$
$$r' = a'\ \mathrm{elqr}(e',g'), \qquad x' = \mathrm{elqr}(e',g') - 1,$$
$$f' = \mathrm{elta}(e',g').$$

Suppose that in the former case the developments of

$$x^a \cos jf, \quad x^a \sin jf, \quad x'^{a'} \cos j'f', \quad x'^{a'} \sin j'f'$$

and in the latter case the developments of

$$\left(\frac{r}{a}\right)^n \cos jf, \quad \left(\frac{r}{a}\right)^n \sin jf, \quad \left(\frac{r'}{a'}\right)^{n'} \cos j'f', \quad \left(\frac{r'}{a'}\right)^{n'} \sin j'f',$$

are

$$\sum [\cos]^i \cos ig, \quad \sum [\sin]^i \sin ig, \quad \sum [\cos]^{i'} \cos i'g', \quad \sum [\sin]^{i'} \sin i'g',$$

where the summations extend from i or $i' = -\infty$ to ∞, and where the coefficients $[\cos]^i$, $[\sin]^i$ satisfy

$$[\cos]^{-i} = [\cos]^i, \qquad [\sin]^{-i} = -[\sin]^i,$$

and in like manner the coefficients $[\cos]^{i'}$, $[\sin]^{i'}$ satisfy

$$[\cos]^{-i'} = [\cos]^{i'}, \qquad [\sin]^{-i'} = -[\sin]^{i'}.$$

It is to be observed that $[\cos]^i$, $[\sin]^i$ are functions of e, and $[\cos]^{i'}$, $[\sin]^{i'}$

[PDF p. 358; printed p. 336.] functions of e'; the accents to the indices i and i' are sufficient to indicate this. Hence observing that

$$\begin{aligned}
\sum[\cos]^i \cos ig \cdot \sum[\cos]^{i'} \cos i'g' &= \sum\sum[\cos]^i\,[\cos]^{i'} \cos(ig+i'g'),\\
\sum[\cos]^i \cos ig \cdot \sum[\sin]^{i'} \sin i'g' &= \sum\sum[\cos]^i\,[\sin]^{i'} \sin(ig+i'g'),\\
\sum[\sin]^i \sin ig \cdot \sum[\cos]^{i'} \cos i'g' &= \sum\sum[\sin]^i\,[\cos]^{i'} \sin(ig+i'g'),\\
\sum[\sin]^i \sin ig \cdot \sum[\sin]^{i'} \sin i'g' &= \sum\sum[\sin]^i\,[\sin]^{i'} \cos(ig+i'g'),
\end{aligned}$$

we have for the products of $x^a\,x'^{a'}$, or as the case may be $\left(\frac{r}{a}\right)^n\left(\frac{r'}{a'}\right)^{n'}$, into

$$\begin{aligned}
&\cos(jf+j'f'), \quad \sum\sum\left([\cos]^i\,[\cos]^{i'} + [\sin]^i\,[\sin]^{i'}\right)\cos(ig+i'g'),\\
&\sin(jf+j'f'), \quad \sum\sum\left([\sin]^i\,[\cos]^{i'} + [\cos]^i\,[\sin]^{i'}\right)\sin(ig+i'g'),
\end{aligned}$$

and thence observing that

$$\begin{aligned}
\cos(j\mathfrak{C}+j'\mathfrak{C}')\sum\sum P^{i,i'}\cos(ig+i'g') &= \sum\sum P^{i,i'}\cos(ig+i'g'+j\mathfrak{C}+j'\mathfrak{C}'),\\
-\sin(j\mathfrak{C}+j'\mathfrak{C}')\sum\sum Q^{i,i'}\sin(ig+i'g') &= \sum\sum Q^{i,i'}\cos(ig+i'g'+j\mathfrak{C}+j'\mathfrak{C}'),
\end{aligned}$$

provided only that $P^{-i,-i'}=P^{i,i'}$, $Q^{-i,-i'}=Q^{i,i'}$, we find for the product of $x^a\,x'^{a'}$ or as the case may be $\left(\frac{r}{a}\right)^n\left(\frac{r'}{a'}\right)^{n'}$ into $\cos(jf+j'f'+j\mathfrak{C}+j'\mathfrak{C}')$ the expression

$$\sum\sum\left([\cos]^i\,[\cos]^{i'} + [\sin]^i\,[\sin]^{i'} + [\sin]^i\,[\cos]^{i'} + [\cos]^i\,[\sin]^{i'}\right)\cos(ig+i'g'+j\mathfrak{C}+j'\mathfrak{C}'),$$

or, finally, the expression

$$\sum\sum\left([\cos]^i + [\sin]^i\right)\left([\cos]^{i'} + [\sin]^{i'}\right)\cos(ig+i'g'+j\mathfrak{C}+j'\mathfrak{C}'),$$

which is the required development of the general term in multiple cosines of the mean anomalies.

VI.

Investigation of the coefficient $C_n(j,j')$.

It is possible that there might be some advantage in developing in the first instance according to the powers of $\cos H$, a process which, as it has been seen, leads very readily to the form of the general term; but the mode which I have adopted is to develope in the first instance according to the powers of η. I put therefore

$$\cos H = \cos(U-U') - 2\eta^2 \sin U \sin U',$$

and we have then for the reciprocal of the distance,

$$\left[r^2 + r'^2 - 2\,r\,r'\cos(U-U') - r\,r'\,(-4\sin U\sin U')\,\eta^2\right]^{-\frac{1}{2}},$$

[PDF p. 359; printed p. 337.] which is to be developed in ascending powers of $\dfrac{1}{r'}$, and we have for the discrete general term of the development,

$$\frac{r^n}{r'^{n+1}}\, C_n(j,j')\,\cos(jU + j'U'),$$

which is the definition of $C_n(j,j')$, the coefficient the value of which is sought for. It has been seen that j, j' are each of them of the same parity with n (even or odd according as n is even or odd), and it has been seen also that it is sufficient to consider the case where j is not negative and not inferior in absolute magnitude to j'.

Expanding in powers of η, and putting as before

$$\Pi a = 1.2.3\ldots a, \qquad \Pi_1(a-\tfrac{1}{2}) = \tfrac{1}{2}.\tfrac{3}{2}\ldots(a-\tfrac{1}{2});$$

the general term is

$$\frac{\Pi_1(\chi-\frac{1}{2})}{\Pi_1\,\Pi_1(-\frac{1}{2})}\cdot\frac{\eta^{2\chi}\,r^{\chi}\,r'^{\chi}}{(r^2+r'^2-2\,r\,r'\cos(U-U'))^{\chi+\frac{1}{2}}}\,(-4\sin U\sin U')^{\chi},$$

where χ extends from 0 to ∞.

The factor $(-4\sin U\sin U')^{\chi}$ consists of a series of multiple cosines, and as usual it is assumed that the cosines to opposite arguments are made to occur with equal coefficients. The form of the general term is $\cos(\lambda U + \lambda' U')$, where λ, λ' have each of them the values χ, $\chi-2$, $\chi-4$,$\ldots -\chi$, that is, λ, λ' are each of them of the same parity with χ. Hence j, j' being as before of the same parity with each other, and ϑ being even or odd according as j, j' and χ are of the same or of opposite parities, the development of $(-4\sin U\sin U')^{\chi}$ will contain a series of terms $\cos\big[(j+\vartheta)U + (j'-\vartheta)U'\big]$, which (since the other factor contains only multiple cosines of $U-U'$) are the only terms which give rise to a term $\cos(jU+j'U')$. I represent the discrete term of $(-4\sin U\sin U')^{\chi}$ which contains the before-mentioned argument by

$$M_{\chi}^{j,j'}\cos\big[(j+\vartheta)U + (j'-\vartheta)U'\big].$$

On the before-mentioned assumption, j not negative and not less in absolute magnitude than j', we have $j+j'$ and $j-j'$ each of them not negative. We must have $j+\vartheta \not> \chi$ and $j'-\vartheta \not> \chi$, that is $\vartheta \not> (\chi - j)$ and $-\vartheta \not> (\chi - j')$, consequently

$$\vartheta \not> \chi - j, \qquad \vartheta \not< -(\chi - j'),$$

or $j'-j \not< -(\chi-j')$ or $2\chi \not< j+j'$. And this relation being assumed, ϑ extends from the inferior limit $-(\chi-j')$ to the superior limit $(\chi-j)$ by steps of two units, the extreme terms being

$$\vartheta = -(\chi - j'), \qquad M_{\chi}^{j,j'}\cos\big[(j+j'-\chi)U + \chi U'\big],$$

the coefficient of U increasing from $j+j'-\chi$ to χ, and that of U' diminishing from χ to $j+j'-\chi$ by steps of two units.

[PDF p. 360; printed p. 338.] Now expanding in ascending powers of $\dfrac{1}{r'}$, write

$$r^{\chi}\, r'^{\chi}\left\{r^2 + r'^2 - 2\,r\,r'\cos(U-U')\right\}^{-\chi-\frac{1}{2}} = \sum R_{\chi}^{i}\cos i(U-U'),$$

where i extends from $-\infty$ to $+\infty$, and $R_{\chi}^{-i} = R_{\chi}^{i}$; so that $R_{\chi}^{i}\cos i(U-U')$ is the discrete general term of the development. The term $R_{\chi}^{i}\cos i(U-U')$ in combination with the term $M_{\chi}^{j,j'}\cos[(j+\vartheta)U + (j'-\vartheta)U']$, gives rise to $R_{\chi}^{i}\,M_{\chi}^{j,j'}$; and restoring the multiplier which has been omitted, and giving to ϑ and χ the different admissible values, we find for the discrete general term containing $\cos(jU + j'U')$ the value

$$\frac{\Pi_1(\chi-\frac{1}{2})}{\Pi_1\,\Pi_1(-\frac{1}{2})}\,\eta^{2\chi}\sum\sum R_{\chi}^{i}\,M_{\chi}^{j,j'}\cos(jU + j'U'),$$

where ϑ extends from the inferior limit $\vartheta = -(\chi - j')$ to the superior limit $\vartheta = \chi - j$, by steps of two units, and χ extends from $\chi = \frac{1}{2}(j+j')$ to $\chi = \infty$. The portion of this containing $\dfrac{r^n}{r'^{n+1}}\cos(jU + j'U')$ is

$$\frac{r^n}{r'^{n+1}}\,C_n(j,j')\cos(jU + j'U'),$$

and we have therefore

$$C_n(j,j') = \sum\sum \frac{\Pi_1(\chi-\frac{1}{2})}{\Pi_1\,\Pi_1(-\frac{1}{2})}\,\eta^{2\chi}\,M_{\chi}^{j,j'}\,R_{\chi}^{n}.$$

There is some speciality in the case $j + j' = 0$, but the result just obtained subsists without variation. To find $M_{\chi}^{j,j'}$ we have

$$M_{\chi}^{j,j'} = \text{disct. coeff. } \cos\bigl[(j+\vartheta)U + (j'-\vartheta)U'\bigr] \text{ in } (-4\sin U\sin U')^{\chi},$$

and putting $\sin U = \dfrac{1}{2i}\bigl(v - \dfrac{1}{v}\bigr)$, $\sin U' = \dfrac{1}{2i}\bigl(v' - \dfrac{1}{v'}\bigr)$ where $i = \sqrt{-1}$ we have

$$(-4\sin U\sin U')^{\chi} = \Bigl(v - \frac{1}{v}\Bigr)^{\chi}\Bigl(v' - \frac{1}{v'}\Bigr)^{\chi},$$

and the function on the right hand contains the term

$$\frac{\Pi\chi}{\Pi f\,\Pi(\chi-f)}\,(-)^{f}\,\frac{\Pi\chi}{\Pi f'\,\Pi(\chi-f')}\,(-)^{f'}\,v^{\chi-2f}\,v'^{\chi-2f'},$$

or putting

$$\chi - 2f = j + \vartheta, \qquad \chi - 2f' = j' - \vartheta$$

and therefore

$$f = \tfrac{1}{2}(\chi - j - \vartheta), \qquad f' = \tfrac{1}{2}(\chi - j' + \vartheta),$$

which give integer values for f, f' since ϑ is even or odd according as j, j', and χ are of the same parity or of opposite parities, and replacing $v^{\chi-2f}\,v'^{\chi-2f'}$ by the half of its value $2\cos\bigl[(j+\vartheta)U + (j'-\vartheta)U'\bigr]$, the term is

$$\frac{2\,\Pi\chi\,\Pi\chi}{\Pi\bigl(\frac{1}{2}(\chi-j-\vartheta)\bigr)\,\Pi\bigl(\frac{1}{2}(\chi+j+\vartheta)\bigr)\,\Pi\bigl(\frac{1}{2}(\chi-j'+\vartheta)\bigr)\,\Pi\bigl(\frac{1}{2}(\chi+j'-\vartheta)\bigr)}\,(-)^{\chi-\vartheta-\frac{1}{2}(j-j')},$$

[PDF p. 361; printed p. 339.] and consequently,

$$M_\chi^{j,j'} = \frac{(-)^{-\chi-\frac{1}{2}(j-j')+\vartheta}\cdot 2\,(\Pi\chi)^2}{\Pi\big(\frac{1}{2}(\chi-j-\vartheta)\big)\,\Pi\big(\frac{1}{2}(\chi+j+\vartheta)\big)\,\Pi\big(\frac{1}{2}(\chi-j'+\vartheta)\big)\,\Pi\big(\frac{1}{2}(\chi+j'-\vartheta)\big)},$$

which is the expression for $M_\chi^{j,j'}$.

The expression for R_χ^i is found in a similar manner, viz., by substituting for $\cos(U-U')$ its exponential expression, by which means the function

$$r^\chi\, r'^\chi \left\{r^2+r'^2-2\,r\,r'\cos(U-U')\right\}^{-\chi-\frac{1}{2}}$$

breaks up into a pair of factors, each of which can be separately expanded; the result, i being positive, or zero, is

$$R_\chi^i = \sum \frac{r^{\chi+i+m}}{r'^{\chi+i+m+1}}\,\frac{\Pi_1(\chi+i+m-\frac{1}{2})\,\Pi_1(\chi+m-\frac{1}{2})}{\Pi_1(-\frac{1}{2})\,\Pi_1(i+m)\,\Pi(\chi-1)\,\Pi m\,\Pi(\chi-1)\,\Pi(\chi-m)}\left(\frac{r}{r'}\right)^{-2m-1}$$

from $m=0$ to $m=\infty$: this may also be written

$$R_\chi^i = \sum \frac{1}{r'^{n+1}}\,r^n\cdot\frac{\Pi_1(\chi-1)\,\Pi_1(\chi+m-\frac{1}{2})}{\Pi_1(-\frac{1}{2})\,\Pi_1(i+m)\,\Pi(\chi-1)\,\Pi m\,\Pi(\chi-m-1)},$$

writing now ϑ for i, and $\chi+\vartheta+2m=n$, that is $m=\frac{1}{2}(n-\chi-\vartheta)$ (n is of the same parity with j, j', and ϑ is even or odd according as j, j' and χ are of the same or opposite parities, m is therefore, as it should be, an integer), R_χ^n contains the term

$$\frac{r^n}{r'^{n+1}}\,\frac{\Pi_1(\frac{1}{2}(n+\vartheta)-\frac{1}{2})\,\Pi_1(\frac{1}{2}(n-\vartheta)-\frac{1}{2})}{\Pi(\frac{1}{2}(n+\vartheta)-\chi)\,\Pi_1(\frac{1}{2}(n-\vartheta)-\chi)\,\Pi(\chi-\frac{1}{2})\,\Pi(\chi-1)}\,K_\chi^{n,\vartheta},$$

if for shortness

$$K_\chi^{n,\vartheta} = \frac{\Pi_1\big(\frac{1}{2}(n+\vartheta)-\frac{1}{2}\big)\,\Pi_1\big(\frac{1}{2}(n-\vartheta)-\frac{1}{2}\big)}{\Pi\big(\frac{1}{2}(n-\chi+\vartheta)\big)\,\Pi\big(\frac{1}{2}(n-\chi-\vartheta)\big)\,\Pi(\chi-1)},$$

a formula which I assume to subsist as well for negative as for positive values of ϑ, so that $K_\chi^{n,-\vartheta}=K_\chi^{n,\vartheta}$. It is to be noticed that if $\chi>n$, then either $n-\chi+\vartheta$ or $n-\chi-\vartheta$ vanishes, and we have therefore $K_\chi^{n,\vartheta}=0$.

Substituting for R_χ^n its value we have

$$C_n(j,j') = \sum\sum \frac{\Pi_1(\chi-\frac{1}{2})}{\Pi_1\,\Pi_1(-\frac{1}{2})}\,\eta^{2\chi}\,M_\chi^{j,j'}\,K_\chi^{n,\vartheta},$$

where $M_\chi^{j,j'}$, $K_\chi^{n,\vartheta}$ have the values already obtained, and as before ϑ extends to $\vartheta=-(\chi-j')$ to $\vartheta=(\chi-j)$ by steps of two units, and χ (since $K_\chi^{n,\vartheta}$ vanishes for $\chi<n$) extends from $\chi=\frac{1}{2}(j+j')$ to $\chi=n$.

[PDF p. 362; printed p. 340.] To simplify; write $\chi = \frac{1}{2}(j+j') + u$, $\vartheta = -(\chi - j') + 2s$, $(\chi - j) - 2t$, so that $s + t = u$, s and t being any integer values (zero not excluded) which satisfy this relation, and lastly u extends from $u = 0$ to $u = n - \frac{1}{2}(j+j')$. We have

$$C_n(j,j') = \sum 2^{j+j'} \frac{\Pi_1\big(\frac{1}{2}(j+j') + u - \frac{1}{2}\big)}{\Pi_1\,\Pi_1(-\frac{1}{2})} \eta^{2(\frac{1}{2}(j+j')+u)} M_\chi^{j,j'} K_\chi^{n,\vartheta},$$

where

$$M_\chi^{j,j'} = (-)^u \frac{\Pi_1\big(\frac{1}{2}(j+j')+u\big)\,\Pi_1\big(\frac{1}{2}(j+j')+u\big)}{\Pi(s)\,\Pi(t)\,\Pi(j+s)\,\Pi(j'+t)},$$

$$K_\chi^{n,\vartheta} = \frac{\Pi_1\big(\frac{1}{2}(n+j)+s-\frac{1}{2}\big)\,\Pi_1\big(\frac{1}{2}(n+j')+t-\frac{1}{2}\big)}{\Pi_1\big(\frac{1}{2}(n-j)-s\big)\,\Pi_1\big(\frac{1}{2}(n-j')-t\big)\,\Pi\big(\frac{1}{2}(j+j')+u-1\big)},$$

and substituting these values, and observing that the result contains a factor $\dfrac{1}{\Pi\big(\frac{1}{2}(j+j')+u\big)\,\Pi_1\big(\frac{1}{2}(j+ \ldots}$ which may be replaced by $\dfrac{2^{j+j'+2u}}{\Pi(j+j'+2u)}$, the result is

$$C_n(j,j') = 2^{j+j'}\,\eta^{j+j'} \sum \frac{\Pi_1\big(\frac{1}{2}(j+j')+u-\frac{1}{2}\big)\,\Pi_1\big(\frac{1}{2}(j+j')+u\big)}{\Pi_1\big(\frac{1}{2}(j+j')+u\big)\,\Pi_1\big(\frac{1}{2}(j+j')+u-\frac{1}{2}\big)}$$
$$\times \frac{\Pi_1\big(\frac{1}{2}(n+j)+s-\frac{1}{2}\big)\,\Pi_1\big(\frac{1}{2}(n+j')+t-\frac{1}{2}\big)}{\Pi(s)\,\Pi(j+s)\,\Pi_1\big(\frac{1}{2}(n-j)-s\big)\,\Pi(t)\,\Pi(j'+t)\,\Pi_1\big(\frac{1}{2}(n-j')-t\big)}\,\eta^{2u},$$

which is easily transformed into

$$C_n(j,j') = 2^{j+j'}\,\eta^{j+j'}\,\frac{\Pi_1\big(\frac{1}{2}(n+j)-\frac{1}{2}\big)\,\Pi_1\big(\frac{1}{2}(n+j')-\frac{1}{2}\big)}{\Pi(j+j')}\,S,$$

if for shortness

$$S = \sum \frac{(-)^{s+t}\,\eta^{2(s+t)}\,(j+j')!}{[\frac{1}{2}(j+j')+s-\frac{1}{2}]^{s+t}\,[\frac{1}{2}(j+j')+t-\frac{1}{2}]^{s+t}\,\Pi(s)\,\Pi(t)\,\Pi(j+s)\,\Pi(j'+t)},$$

$$s + t = u, \quad u = 0 \text{ to } u = n - \tfrac{1}{2}(j+j');$$

the value of the sum S is

$$S = (1-\eta^2)^{\frac{1}{2}(j-j')}\,F\big(-n+j,\ n+j+1,\ j+j'+1,\ \eta^2\big),$$

and we have consequently

$$C_n(j,j') = 2^{j+j'}\,\eta^{j+j'}\,\frac{\Pi_1\big(\frac{1}{2}(n+j)-\frac{1}{2}\big)\,\Pi_1\big(\frac{1}{2}(n+j')-\frac{1}{2}\big)}{\Pi_1\big(\frac{1}{2}(n-j)\big)\,\Pi_1\big(\frac{1}{2}(n-j')\big)\,\Pi(j+j')}\,(1-\eta^2)^{\frac{1}{2}(j-j')}\,F\big(-n+j,\ n+j+1,\ j+j'+1,\ \eta^2\big)$$

the required expression for $C_n(j,j')$. It only remains to prove the formula for S.

[PDF p. 363; printed p. 341.] For this purpose, observing the equation $s+t=u$, I form the equations

$$\frac{1}{(j+j'+2u)^u} = \frac{1}{[\frac{1}{2}(j+j')+u-\frac{1}{2}]^u[\frac{1}{2}(j+j')+u-1]^u},$$

$$[\tfrac{1}{2}(j+j')+u]^u = [\tfrac{1}{2}(j+j')+s]^u\,[\tfrac{1}{2}(j+j')+t]^u,$$

$$[\tfrac{1}{2}(j+j')+u]^u = [\tfrac{1}{2}(j+j')+s]^u\,[\tfrac{1}{2}(j+j')+t]^u,$$

and thence

$$\frac{2^u}{(j+j'+2u)^u} = \frac{[\frac{1}{2}(j+j')+s]^u\,[\frac{1}{2}(j+j')+t]^u}{[\frac{1}{2}(j+j')+s-\frac{1}{2}]^u\,[\frac{1}{2}(j+j')+t-\frac{1}{2}]^u},$$

and it is proper to remark that the summation as regards u may be continued indefinitely; for, if u exceed $n-\frac{1}{2}(j+j')$, then one at least of the relations $s>\frac{1}{2}(n-j)$, $t>\frac{1}{2}(n-j')$ must hold good, and at least one of the factorials $[\frac{1}{2}(n-j)]^s$, $[\frac{1}{2}(n-j')]^t$ will vanish. The two factors in the second sum are the general terms of two hypergeometric series. In fact, if we put

$$\alpha=-\tfrac{1}{2}n+\tfrac{1}{2}j, \qquad \beta=-\tfrac{1}{2}n+\tfrac{1}{2}j', \qquad \epsilon=\tfrac{1}{2}j+\tfrac{1}{2}j'+\tfrac{1}{2},$$

and therefore

$$\epsilon-\alpha=\tfrac{1}{2}n+\tfrac{1}{2}j'+\tfrac{1}{2}, \qquad \epsilon-\beta=\tfrac{1}{2}n+\tfrac{1}{2}j+\tfrac{1}{2},$$

and if, for shortness, we use $\{a\}^s$ to denote the factorial $a(a+1)\ldots(a+s-1)$ of the increment positive unity, then we have

$$\alpha\sum\frac{\{\alpha\}^s\,\{\beta\}^s}{\{\epsilon+\frac{1}{2}\}^s\Pi s}\eta^{2s} = \sum\frac{1}{\Pi s\,\{\epsilon+\frac{1}{2}\}^s}\,\frac{\Pi(j+s)}{\Pi j}\cdot\frac{1}{\Pi(s)}\cdot\eta^{2s},$$

the two hypergeometric series being consequently

$$1+\frac{\alpha\cdot\beta}{1\cdot\epsilon}+\frac{\alpha\cdot\alpha+1}{1\cdot 2}\cdot\frac{\beta\cdot\beta+1}{\epsilon\cdot\epsilon+1}\,\eta^4+\&\text{c.}$$

[PDF p. 364; printed p. 342.] I assume the truth of the following remarkable theorem, [see 211] viz.:

"The series formed from the product of the two hypergeometric series,

$$F(\alpha,\, \beta,\, \epsilon+\tfrac{1}{2},\, \eta^2)\, F(\epsilon-\alpha,\, \epsilon-\beta,\, \epsilon+\tfrac{1}{2},\, \eta^2),$$

by multiplying the successive terms of the product by 1, $\dfrac{1}{2}$, $\dfrac{1\cdot 3}{2\cdot 4}$, &c. respectively, is

$$= (1-\eta^2)^{\alpha+\beta-\epsilon}\, F(2\alpha,\, 2\beta,\, 2\epsilon,\, \eta^2)."$$

Hence, observing that the general term of S is formed precisely in the manner in question, we have

$$S = (1-\eta^2)^{\alpha+\beta-\epsilon}\, F(2\alpha,\, 2\beta,\, 2\epsilon,\, \eta^2)$$

or, substituting for α, β, ϵ their values

$$S = (1-\eta^2)^{-\frac{1}{2}(n+\frac{1}{2})}\, F(-n+j,\, -n+j',\, j+j'+1,\, \eta^2),$$

which is the required value.

It is clear that α, β are interchangeable with $\epsilon-\alpha$, $\epsilon-\beta$; that is, we have

$$F(2\alpha,\, 2\beta,\, 2\epsilon,\, \eta^2) = F(2\epsilon-2\alpha,\, 2\epsilon-2\beta,\, 2\epsilon,\, \eta^2)$$

or

$$F(2\epsilon-2\alpha,\, 2\epsilon-2\beta,\, 2\epsilon,\, \eta^2) = (1-\eta^2)^{\alpha+2\beta-\epsilon-\epsilon}\, F(2\alpha,\, 2\beta,\, 2\epsilon,\, \eta^2),$$

which is a known property of hypergeometric series. The form

$$S = (1-\eta^2)^{-\frac{1}{2}(n+\frac{1}{2})}\, F\big(-n+j,\, n+j+1,\, j+j'+1,\, \eta^2\big)$$

is obviously less convenient than the one above mentioned, since the new expression is encumbered by a denominator $(1-\eta^2)^{\frac{1}{2}(j-j')}$, which really divides out, the finite hypergeometric series containing as a factor the square of such denominator. I have only noticed this for the verification which it affords by showing that j, j' may be interchanged.

VII.

The above expression of $C_n(j,j')$ is not, in its actual form, given in Hansen's memoir. The comparison with Hansen's formula is as follows: The formula (36), p. 329, is

$$C(n-2f',\, -(n-2f-2(f)) = H\cos^n \tfrac{1}{2}J\, \tan^{2f}\tfrac{1}{2}J\, F\big(-(2n-2f-2(f)),\, -2f,\, 2f+1,\, -\tan^2\tfrac{1}{2}J\big)$$

where

$$H = 2^{-n}\, \Pi(n-f)\, \Pi(n-f-(f))\, \Pi(f+(f))\, \Pi 2(f)$$

[PDF p. 365; printed p. 343.] and the formula (37), p. 330, is

$$F\big(-(2n-2f-2(f)),\, -2f,\, 2(f)+1,\, -\tan^2\tfrac{1}{2}J\big)$$

$$= \cos^{-4n+4f+4(f)}\tfrac{1}{2}J\, F\big(-(2n-2f-2(f)),\, 2f+2(f)+1,\, 2(f)+1,\, \sin^2\tfrac{1}{2}J\big).$$

Combining these, and putting $\sin\frac{1}{2}J = \eta$, we have

$$C\big(n-2f,\, -(n-2f-2(f))\big) = H\eta^{2(f)}\,(1-\eta^2)^{-n+f+(f)}\,F\big(-(2n-2f-2(f)),\, 2f+2(f)+1,\, 2(f)+1,\, \eta^2\big),$$

and then, putting

$$n-2f = j', \qquad -n+2f+2(f) = j,$$

and for $C(j', j)$, $= C(j, j')$, writing $C_n(j, j')$, we have

$$C_n(j,j') = H\eta^{j+j'}\,(1-\eta^2)^{\frac{1}{2}(j-j')}\,F\big(-n+j,\, n+j+1,\, j+j'+1,\, \eta^2\big)$$

where

$$H = \frac{1}{2^n\,\Pi(n+j')\,\Pi(n-j')\,\Pi\frac{1}{2}(n+j)\,\Pi\frac{1}{2}(n-j)\,\Pi(j+j')},$$

or, finally, since

$$\Pi(n+j) = 2^{n+j}\,\Pi_1\tfrac{1}{2}(n+j)\,\Pi_1\big(\tfrac{1}{2}(n+j)-\tfrac{1}{2}\big),$$

$$\Pi(n+j') = 2^{n+j'}\,\Pi_1\tfrac{1}{2}(n+j')\,\Pi_1\big(\tfrac{1}{2}(n+j')-\tfrac{1}{2}\big),$$

we find

$$C_n(j,j') = 2^{j+j'}\,\frac{\Pi_1\big(\frac{1}{2}(n+j)-\frac{1}{2}\big)\,\Pi_1\big(\frac{1}{2}(n+j')-\frac{1}{2}\big)}{\Pi_1\frac{1}{2}(n-j)\,\Pi_1\frac{1}{2}(n-j')\,\Pi(j+j')}\,\eta^{j+j'}\,(1-\eta^2)^{\frac{1}{2}(j-j')}\,F\big(-n+j,\, n+j+1,\, j+j'+1,\, \eta^2\big),$$

which is the formula of the present memoir.

215. A Supplementary Memoir on the Problem of Disturbed Elliptic Motion

[PDF p. 366; printed p. 344.]

[From the *Memoirs of the Royal Astronomical Society*, vol. XXVIII. pp. 217–234. Read January 12, 1860.]

The present memoir contains an investigation upon fundamentally different principles of a system of formula given in my "Memoir on the Problem of Disturbed Elliptic Motion," *Ast. Soc. Mem.* t. XXVII. pp. 1–29 (1858), [212]; it is shown how the formulae are deduced by means of the general equations of the *Mécanique Analytique*, from the expression for the *Vis viva* function T, in terms of the coordinates (the radius vector, the departure) of the body in its instantaneous orbit, viz., the ultimate form of this function is $T\,dt^2 = \frac{1}{2}(dr^2 + r^2\,d\phi^2)$, but T contains in the first instance terms, not identically vanishing, but which are to be equated to zero, thus furnishing equations of the problem which could not be obtained from the foregoing ultimate form of T. The investigation throws, I think, a further light upon the system of formulae, and completes the development which I was anxious to give of the dynamical problem. I have been a great deal indebted, in the composition of the memoir, to a correspondence some time ago with Professor Donkin on the general subject.

The word *orbit* is used to denote a great circle of the sphere, and it is assumed that in any orbit there is an origin of longitudes; the angular position of a body in reference to the orbit is determined by the longitude and latitude. It is ultimately assumed that the longitude is measured from an origin which is what I have called a *departure-point*; viz., in the general case of a variable orbit the departure-point is the point of intersection of the orbit by any orthogonal trajectory of the successive positions of the orbit: this includes the case of a fixed orbit, where the departure-point is simply a fixed point. As regards points in the orbit, the word *departure* may be used instead of longitude. In the present memoir the origin is in the first instance taken to be, not a departure-point, but an arbitrarily varying point of the orbit.

[PDF p. 367; printed p. 345.] The mutual position of any two orbits whatever, say the position of an orbit $x'y'$ and of the origin x' therein, in reference to the orbit xy and the origin x therein, is determined by

θ, the longitude of node,
ϖ, the departure of node,
ϕ, the inclination,

where the expression *departure of node* is used by way of distinction from longitude of node, the departure referring to the orbit $x'y'$ and origin x', the positions of which are to be determined, and the longitude to the orbit of reference xy and origin x. And this distinction will be preserved throughout. It should be recollected that it is not as yet assumed that the origins are departure-points.

Consider now a fixed orbit x_0y_0 and fixed origin x_0 therein, and suppose that the orbit x_1y_1 and origin x_1 therein are determined in reference to the orbit x_0y_0 and origin x_0 therein, by

θ', the longitude of node,
ϖ', the departure of node,
ϕ', the inclination,

the orbit x_2y_2 and origin x_2 therein, in reference to the orbit x_1y_1 and origin x_1 therein, by

Θ, the longitude of node,
$\mathit{\Sigma}$, the departure of node,
$\mathit{\Phi}$, the inclination,

and, finally, the position of the body in reference to the orbit x_2y_2 and origin x_2 therein, by

r, the radius vector,
v, the longitude,
y, the latitude.

It is required to find the expression of T, the *Vis viva* function, or half square of the velocity. We may imagine the rectangular axes $x_0y_0z_0$, $x_1y_1z_1$, $x_2y_2z_2$, the positions of which are determined as above, and the rectangular axes $x_3y_3z_3$, that of x_3 passing through the body, that of y_3 lying in the orbit x_2y_2, 90° in advance of the body or

[PDF p. 368; printed p. 346.] passing through the pole of the latitude circle, and that of z_3 in the latitude circle 90° above the body. Or considering x_3, y_3, z_3 as points of the sphere, their positions in reference to the orbit x_2y_2 and origin x_2 therein are determined as follows, viz.,

$$\begin{array}{lcr} \text{for } x_3, & \text{longitude is } v, & \text{and latitude } y, \\ y_3, & v+90^\circ, & 0, \\ z_3, & v, & y+90^\circ, \end{array}$$

and we may suppose that (x_0, y_0, z_0), (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3), are the coordinates of any point in reference to these sets of axes respectively, the coordinates of the body in reference to the axes $x_3y_3z_3$ being obviously $(r, 0, 0)$.

The displacements in the directions of the axes $x_3y_3z_3$ respectively of a point the coordinates of which are x_3, y_3, z_3, are as usual

$$\begin{aligned} &dx_3 + (-R\,y_3 + Q\,z_3)\,dt, \\ &dy_3 + (-P\,z_3 + R\,x_3)\,dt, \\ &dz_3 + (-Q\,x_3 + P\,y_3)\,dt, \end{aligned}$$

where the expressions of the rotations P, Q, R are as follows, viz.,

$$\begin{aligned} P\,dt = {} & \sin y\,dv \\ & + \cos y\big[\sin(v-\varSigma)\sin\varPhi\,d\Theta + \cos(v-\varSigma)\,d\varPhi\big] \\ & + \sin y\big[-d\varSigma + \cos\varPhi\,d\Theta\big] \\ & + \big\{\cos y\cos(v-\varSigma)\big\}\big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big] \\ & + \big\{\cos y\sin(v-\varSigma)\cos\varPhi - \sin y\sin\varPhi\big\}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big] \\ & + \big\{\cos y\sin(v-\varSigma)\sin\varPhi + \sin y\cos\varPhi\big\}\big[-d\varpi' + \cos\phi'\,d\theta'\big], \end{aligned}$$

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$$\begin{aligned} Q\,dt = dy &\\ &+ \bigl[\cos(v-\mathit{\Sigma})\sin\mathit{\Phi}\,d\Theta - \sin(v-\mathit{\Sigma})\,d\mathit{\Phi}\bigr]\\ &- \sin(v-\mathit{\Sigma})\bigl[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\bigr]\\ &+ \cos(v-\mathit{\Sigma})\cos\mathit{\Phi}\bigl[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\bigr]\\ &+ \cos(v-\mathit{\Sigma})\sin\mathit{\Phi}\bigl[-d\varpi' + \cos\phi'\,d\theta'\bigr], \end{aligned}$$

$$\begin{aligned} R\,dt = \cos y\,dv &\\ &- \sin y\bigl[\sin(v-\mathit{\Sigma})\sin\mathit{\Phi}\,d\Theta + \cos(v-\mathit{\Sigma})\,d\mathit{\Phi}\bigr]\\ &+ \cos y\bigl[-d\mathit{\Sigma} + \cos\mathit{\Phi}\,d\Theta\bigr]\\ &+ \bigl\{-\sin y\cos(v-\mathit{\Sigma})\bigr\}\bigl[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\bigr]\\ &+ \bigl\{-\sin y\sin(v-\mathit{\Sigma})\cos\mathit{\Phi} - \cos y\sin\mathit{\Phi}\bigr\}\bigl[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\bigr]\\ &+ \bigl\{-\sin y\sin(v-\mathit{\Sigma})\sin\mathit{\Phi} + \cos y\cos\mathit{\Phi}\bigr\}\bigl[-d\varpi' + \cos\phi'\,d\theta'\bigr]. \end{aligned}$$

For the body itself we must write r, 0, 0, in the place of x_3, y_3, z_3; the displacements are therefore

$$dr, \quad rR\,dt, \quad rQ\,dt;$$

and the expression for the *Vis viva* function is

$$T\,dt^2 = \tfrac{1}{2}\bigl(dr^2 + r^2(Q^2+R^2)\,dt^2\bigr),$$

where Q, R have the above-mentioned values.

Let Ω be a function of the coordinates x_0, y_0, z_0 of a point, Ω will be a function of x_3, y_3, z_3, and also of the quantities θ', Θ, $\mathit{\Sigma}$, ϖ', ϕ', $\mathit{\Phi}$, which determine the positions of the axes $x_3y_3z_3$, and the complete differential of Ω will be

$$d\Omega = \frac{d\Omega}{dx_3}\bigl[dx_3+(-R\,y_3+Q\,z_3)\,dt\bigr]+\frac{d\Omega}{dy_3}\bigl[dy_3+(-P\,z_3+R\,x_3)\,dt\bigr]+\frac{d\Omega}{dz_3}\bigl[dz_3+(-Q\,x_3+P\,y_3)\,dt\bigr].$$

I assume for the present the truth of the following proposition, viz., that when Ω is a function of the coordinates of the body (in which case, as before, the values of x_3, y_3, z_3 are r, 0, 0), we have

$$\frac{d\Omega}{dx_3} = \frac{d\Omega}{dr}, \qquad \frac{d\Omega}{dy_3} = \frac{1}{r}\sec y\,\frac{d\Omega}{dv}, \qquad \frac{d\Omega}{dz_3} = \frac{1}{r}\,\frac{d\Omega}{dy}.$$

[PDF p. 370; printed p. 348.] Ω is here a function of r, v, y, θ', Θ, $\varSigma$, ϖ', ϕ', $\varPhi$, or, as we may express it, $\Omega = \Omega(r, v, y, \theta', \Theta, \varSigma, \varpi', \phi', \varPhi)$; and the expression for the complete differential, by what precedes, is

$$d\Omega = \frac{d\Omega}{dr}\,dr + R\,dt\sec y\,\frac{d\Omega}{dv} + Q\,dt\,\frac{d\Omega}{dy},$$

or, substituting for Q, R, their values, the expression is

$$\begin{aligned}
d\Omega = {} & \frac{d\Omega}{dr}\,dr \\
& + \sec y\,\frac{d\Omega}{dv}\Big\{\cos y\,dv - \sin y\big[\sin(v-\varSigma)\sin\varPhi\,d\Theta + \cos(v-\varSigma)\,d\varPhi\big] \\
& \qquad + \cos y\big[-d\varSigma + \cos\varPhi\,d\Theta\big] \\
& \qquad + \{-\sin y\cos(v-\varSigma)\}\big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big] \\
& \qquad + \{-\sin y\sin(v-\varSigma)\cos\varPhi - \cos y\sin\varPhi\}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big] \\
& \qquad + \{-\sin y\sin(v-\varSigma)\sin\varPhi + \cos y\cos\varPhi\}\big[-d\varpi' + \cos\phi'\,d\theta'\big]\Big\} \\
& + \frac{d\Omega}{dy}\Big\{dy + \big[\cos(v-\varSigma)\sin\varPhi\,d\Theta - \sin(v-\varSigma)\,d\varPhi\big] \\
& \qquad - \sin(v-\varSigma)\big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big] \\
& \qquad + \cos(v-\varSigma)\cos\varPhi\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big] \\
& \qquad + \cos(v-\varSigma)\sin\varPhi\big[-d\varpi' + \cos\phi'\,d\theta'\big]\Big\}
\end{aligned}$$

where, on the right-hand side, the terms involving the differentials of r, v, y, are, as they should be, $\frac{d\Omega}{dr}\,dr + \frac{d\Omega}{dv}\,dv + \frac{d\Omega}{dy}\,dy$.

I have given the complete expression, as it may be useful; but for the present purpose it will be sufficient to attend to the terms involving $\varSigma$, Θ, $\varPhi$. We have

$$\frac{d\Omega}{d\varSigma}+\frac{d\Omega}{d\Theta}+\frac{d\Omega}{d\varPhi} = \sec y\,\frac{d\Omega}{dv}\Big(-\sin y\sin(v-\varSigma)\sin\varPhi - \cos y\Big) + \frac{d\Omega}{dy}\big\{\cos(v-\varSigma)\sin\varPhi\,d\Theta - \sin(v-\varSigma)\,d\varPhi\big\},$$

and consequently

[PDF p. 371; printed p. 349.]

$$\begin{aligned}
\frac{d\Omega}{d\varSigma} &= -\frac{d\Omega}{dv},\\
\frac{d\Omega}{d\Theta} &= \left\{\cos\varPhi - \tan y\sin(v-\varSigma)\sin\varPhi\right\}\frac{d\Omega}{dv} - \cos(v-\varSigma)\sin\varPhi\,\frac{d\Omega}{dy},\\
\frac{d\Omega}{d\varPhi} &= -\tan y\cos(v-\varSigma)\,\frac{d\Omega}{dv} + \sin(v-\varSigma)\,\frac{d\Omega}{dy};
\end{aligned}$$

which, it is to be observed, are partial differential equations satisfied by Ω considered as a function of the form $\Omega = \Omega(r, v, y, \Theta, \varSigma, \varPhi)$; Ω is, in fact, a function of three quantities only (say the coordinates x_1, y_1, z_1); and it is clear, a priori, that when thus expressed as a function of six quantities, it must satisfy three partial differential equations.

I write in the second equation $\dfrac{1}{\sin\varPhi}\dfrac{d\Omega}{d\Theta}$ for $\dfrac{d\Omega}{d\Theta}$, and I then put $y = 0$; we thus have

$$\cos(v-\varSigma)\,\frac{d\Omega}{dy} = \cot\varPhi\,\frac{d\Omega}{dv} - \operatorname{cosec}\varPhi\,\frac{d\Omega}{d\Theta},$$

$$\sin(v-\varSigma)\,\frac{d\Omega}{dy} = \frac{d\Omega}{d\varPhi}.$$

I propose, as already mentioned, to apply the formulae to the demonstration of the results obtained in my Memoir on the Problem of Disturbed Elliptic Motion. It will be remembered that x_0y_0 and x_0 denote a fixed orbit and fixed origin therein, x_1y_1 and x_1 an arbitrarily varying orbit and origin therein. I assume, however, that x_1 is a departure-point, so that we have

$$d\varpi' + \cos\phi'\,d\theta' = 0.$$

The orbit x_2y_2 will be taken to be the varying instantaneous orbit of the body itself, that is, we have constantly $y = 0$; and it is assumed that x_2 is a departure-point in this orbit. To conform to my former notation, I write $\flat$ instead of v; the position of the body in the varying instantaneous orbit is consequently determined by

r, the radius vector,

$\flat$, the departure.

The entire displacement of the body arises from the displacements dr and $r\,d\flat$ in the direction of the radius vector, and perpendicular to this direction, in the instantaneous orbit; that is, we have $Q = 0$, $R = d\flat$, and the expression for the *Vis viva* function is simply

$$T\,dt^2 = \tfrac{1}{2}(dr^2 + r^2\,d\flat^2),$$

and, putting in the foregoing expressions of Q and R, $y = 0$, and $\flat$ in the place of v, the equations $Q = 0$, $R\,dt = d\flat$, give

$$\begin{aligned}
&\cos(\flat-\varSigma)\sin\varPhi\,d\Theta - \sin(\flat-\varSigma)\,d\varPhi =\\
&\quad \sin(\flat-\varSigma)\left[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\right]\\
&\quad - \cos(\flat-\varSigma)\cos\varPhi\left[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\right]\\
&\quad - \sin\varPhi\left[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\right],
\end{aligned}$$

[PDF p. 372; printed p. 350.] Now considering r, $\flat$, y as the coordinates (y being ultimately put equal to zero) the general equations of motion are

$$\frac{d}{dt}\Big(\frac{dT}{dr'}\Big) - \frac{dT}{dr} = \frac{dV}{dr},$$

$$\frac{d}{dt}\Big(\frac{dT}{d\flat'}\Big) - \frac{dT}{d\flat} = \frac{dV}{d\flat},$$

$$\frac{d}{dt}\Big(\frac{dT}{dy'}\Big) - \frac{dT}{dy} = \frac{dV}{dy},$$

if for the moment r', $\flat'$, y' denote the differential coefficients of r, $\flat$, y. The expression of the force function V is $V = \dfrac{n^2a^3}{r} + \Omega$ where n^2a^3 denotes an absolute constant, the sum of the masses, and the disturbing function Ω has a contrary sign to that of the *Mécanique Céleste*. We may in the first and second equations write at once $T\,dt^2 = \frac{1}{2}(dr^2 + r^2\,d\flat^2)$, and these equations thus become

$$\frac{d}{dt}\frac{dr}{dt} - r\Big(\frac{d\flat}{dt}\Big)^2 = -\frac{n^2a^3}{r^2} + \frac{d\Omega}{dr},$$

$$\frac{d}{dt}\Big(r^2\frac{d\flat}{dt}\Big) = \frac{d\Omega}{d\flat}.$$

If in the third equation we take the general value of T and after the differentiation with respect to y', make the substitutions $y = 0$, $Q = 0$, $R\,dt = d\flat$, we find

$$\frac{dT}{dy'} = 0,$$

and the equation thus becomes

$$-\frac{dT}{dy} = \frac{d\Omega}{dy}.$$

But $T\,dt^2 = \frac{1}{2}\big\{dr^2 + r^2(Q^2 + R^2)\,dt^2\big\}$, consequently

$$-\frac{d}{dy}\,T\,dt = -\frac{dT}{dy} = r^2\Big(Q\,\frac{dQ}{dy} + R\,\frac{dR}{dy}\Big)\,dt = r^2\,d\flat\,\frac{dR}{dy},$$

and from the value of R, putting after the differentiation $y = 0$, we find

$$\begin{aligned}\frac{dR}{dy}\,dt = {} & -\sin(\flat - \varSigma)\sin\varPhi\,d\Theta - \cos(\flat - \varSigma)\,d\varPhi \\ & - \cos(\flat - \varSigma)\big[\sin(\Theta - \varpi')\sin\phi'\,d\theta' + \cos(\Theta - \varpi')\,d\phi'\big] \\ & - \sin(\flat - \varSigma)\cos\varPhi\big[\cos(\Theta - \varpi')\sin\phi'\,d\theta' - \sin(\Theta - \varpi')\,d\phi'\big].\end{aligned}$$

[PDF p. 373; printed p. 351.] I write $r^2 \dfrac{d\flat}{dt} = h$. We thus have

$$\frac{d}{dt}\,\frac{dr}{dt} - \left(\frac{d\flat}{dt}\right)^2 r = -\frac{n^2a^3}{r^2} + \frac{d\Omega}{dr},$$

$$r^2\,\frac{d\flat}{dt} = h,$$

and

$$dh = \frac{d\Omega}{d\flat}\,dt;$$

the remaining equations are

$$d\mathit{\Sigma} - \cos\mathit{\Phi}\,d\Theta = -\sin\mathit{\Phi}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big],$$

$$\begin{aligned}\cos(\flat-\mathit{\Sigma})\sin\mathit{\Phi}\,d\Theta - \sin(\flat-\mathit{\Sigma})\,d\mathit{\Phi} = {}&\sin(\flat-\mathit{\Sigma})\big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big]\\ &-\cos(\flat-\mathit{\Sigma})\cos\mathit{\Phi}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big],\end{aligned}$$

$$\begin{aligned}\sin(\flat-\mathit{\Sigma})\sin\mathit{\Phi}\,d\Theta + \cos(\flat-\mathit{\Sigma})\,d\mathit{\Phi} = {}&\frac{1}{h}\,\frac{d\Omega}{dy}\,dt\\ &-\cos(\flat-\mathit{\Sigma})\big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big]\\ &-\sin(\flat-\mathit{\Sigma})\cos\mathit{\Phi}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big].\end{aligned}$$

Moreover we have

$$\cos(\flat-\mathit{\Sigma})\,\frac{d\Omega}{dy} = \cot\mathit{\Phi}\,\frac{d\Omega}{dv} - \operatorname{cosec}\mathit{\Phi}\,\frac{d\Omega}{d\Theta},$$

$$\sin(\flat-\mathit{\Sigma})\,\frac{d\Omega}{dy} = \frac{d\Omega}{d\mathit{\Phi}},$$

by means of which the preceding two equations are reduced to

$$\begin{aligned}d\Theta = {}&-\frac{\operatorname{cosec}\mathit{\Phi}}{h}\,\frac{d\Omega}{d\mathit{\Phi}}\,dt\\ &-\cot\mathit{\Phi}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big],\\ d\mathit{\Phi} = {}&\frac{\cot\mathit{\Phi}}{h}\,\frac{d\Omega}{d\Theta}\,dt - \frac{\operatorname{cosec}\mathit{\Phi}}{h}\,\frac{d\Omega}{dv}\,dt\\ &-\big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big],\end{aligned}$$

and we then have

$$d\mathit{\Sigma} = -\frac{\cot\mathit{\Phi}}{h}\,\frac{d\Omega}{d\mathit{\Phi}}\,dt - \operatorname{cosec}\mathit{\Phi}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big].$$

[PDF p. 374; printed p. 352.] The entire system consequently is

$$\frac{d}{dt}\Big(\frac{dr}{dt}\Big) - r\Big(\frac{d\flat}{dt}\Big)^2 = -\frac{n^2a^3}{r^2} + \frac{d\Omega}{dr},$$

$$r^2\,\frac{d\flat}{dt} = h;$$

and

$$\begin{aligned}
dh &= \frac{d\Omega}{d\flat}\,dt,\\
d\Theta &= -\frac{\operatorname{cosec}\mathit{\Phi}}{h}\,\frac{d\Omega}{d\mathit{\Phi}}\,dt - \cot\mathit{\Phi}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big],\\
d\mathit{\Sigma} &= -\frac{\cot\mathit{\Phi}}{h}\,\frac{d\Omega}{d\mathit{\Phi}}\,dt - \operatorname{cosec}\mathit{\Phi}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big],\\
d\mathit{\Phi} &= \frac{\cot\mathit{\Phi}}{h}\,\frac{d\Omega}{d\Theta}\,dt - \frac{\operatorname{cosec}\mathit{\Phi}}{h}\,\frac{d\Omega}{dv}\,dt - \big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big],
\end{aligned}$$

where $\Omega = \Omega(r, \flat, \Theta, \mathit{\Sigma}, \mathit{\Phi})$. We may add the before-mentioned equation,

$$d\mathit{\Sigma} - \cos\mathit{\Phi}\,d\Theta = -\sin\mathit{\Phi}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big].$$

To obtain the formulae of my Memoir before referred to, it would only be necessary to write $h = na^2\sqrt{1-e^2}$, and complete the solution by applying to the first equation the method of the variation of the elements; but I prefer leaving the system in its present form, as it is one which may perhaps be useful.

The formulae have especial reference to the lunar theory, and the orbit x_0y_0 denotes the fixed ecliptic, x_1y_1 the varying ecliptic, and x_2y_2 the instantaneous orbit of the moon. But if, instead of considering with Hansen the instantaneous orbit of the moon, the position of the moon is determined (as in the theories of Clairaut, Plana, and others) by

r, the radius vector,
v, the longitude,
y, the latitude,

where the longitude is measured on the variable ecliptic, then if, in the general expressions for P, Q, R, we first neglect the terms involving $d\theta'$, $d\varpi'$, $d\phi'$, and then write θ, ϖ, ϕ, in the place of Θ, $\mathit{\Sigma}$, $\mathit{\Phi}$, we find

$$\begin{aligned}
P\,dt &= \sin y\,dv\\
&\qquad + \cos y\big[\sin(v-\varpi)\sin\phi\,d\theta + \cos(v-\varpi)\,d\phi\big]\\
&\qquad + \sin y\big[-d\varpi + \cos\phi\,d\theta\big],\\
Q\,dt &= dy\\
&\qquad + \big[\cos(v-\varpi)\sin\phi\,d\theta - \sin(v-\varpi)\,d\phi\big],
\end{aligned}$$

[PDF p. 375; printed p. 353.]

$$\begin{aligned}R\,dt = \cos y\,dv \\ &- \sin y\bigl[\sin(v-\varpi)\sin\phi\,d\theta + \cos(v-\varpi)\,d\phi\bigr] \\ &+ \cos y\bigl[-d\varpi + \cos\phi\,d\theta\bigr],\end{aligned}$$

and with these values the expression for the *Vis viva* function is, as before,

$$T\,dt^2 = \tfrac{1}{2}\bigl(dr^2 + r^2(Q^2+R^2)\,dt^2\bigr).$$

The longitude may be measured from a departure-point, and then the expressions for P, Q, R, may be simplified by omitting the terms which contain $d\varpi + \cos\phi\,d\theta$.

Addition.

I have in the course of the memoir used some properties connected with the theory of rotation, but it is proper to give the analytical investigation.

Suppose that the rectangular coordinates (X, Y, Z) are connected with the rectangular coordinates $(X_{\prime}, Y_{\prime}, Z_{\prime})$ by the equations

$$\begin{aligned}X &= \alpha\,X_{\prime} + \beta\,Y_{\prime} + \gamma\,Z_{\prime},\\ Y &= \alpha' X_{\prime} + \beta' Y_{\prime} + \gamma' Z_{\prime},\\ Z &= \alpha'' X_{\prime} + \beta'' Y_{\prime} + \gamma'' Z_{\prime},\end{aligned}$$

then we have

$$\begin{aligned}dX &= \alpha\,dX_{\prime} + \beta\,dY_{\prime} + \gamma\,dZ_{\prime} + X_{\prime}\,d\alpha + Y_{\prime}\,d\beta + Z_{\prime}\,d\gamma,\\ dY &= \alpha'\,dX_{\prime} + \beta'\,dY_{\prime} + \gamma'\,dZ_{\prime} + X_{\prime}\,d\alpha' + Y_{\prime}\,d\beta' + Z_{\prime}\,d\gamma',\\ dZ &= \alpha''\,dX_{\prime} + \beta''\,dY_{\prime} + \gamma''\,dZ_{\prime} + X_{\prime}\,d\alpha'' + Y_{\prime}\,d\beta'' + Z_{\prime}\,d\gamma'',\end{aligned}$$

and then, putting

$$\begin{aligned}p\,dt &= \gamma\,d\beta + \gamma'\,d\beta' + \gamma''\,d\beta'',\\ q\,dt &= \alpha\,d\gamma + \alpha'\,d\gamma' + \alpha''\,d\gamma'',\\ r\,dt &= \beta\,d\alpha + \beta'\,d\alpha' + \beta''\,d\alpha'',\end{aligned}$$

where p, q, r are the rotations round $X_{\prime}$, $Y_{\prime}$, $Z_{\prime}$, from $Y_{\prime}$ to $Z_{\prime}$, $Z_{\prime}$ to $X_{\prime}$, and $X_{\prime}$ to $Y_{\prime}$ respectively, we have

$$\begin{aligned}\alpha\,dX + \alpha'\,dY + \alpha''\,dZ &= dX_{\prime} + (-r\,Y_{\prime} + q\,Z_{\prime})\,dt,\\ \beta\,dX + \beta'\,dY + \beta''\,dZ &= dY_{\prime} + (-p\,Z_{\prime} + r\,X_{\prime})\,dt,\\ \gamma\,dX + \gamma'\,dY + \gamma''\,dZ &= dZ_{\prime} + (-q\,Z_{\prime} + p\,Y_{\prime})\,dt,\end{aligned}$$

where, considering XYZ as fixed axes, the left-hand sides are the displacements in the directions of the moveable axes $X_{\prime}Y_{\prime}Z_{\prime}$ arising from the variations of the coordinates $X_{\prime}Y_{\prime}Z_{\prime}$ and the variation of the position of these axes; and these displace-

[PDF p. 376; printed p. 354.] ments have therefore the values given by the right-hand sides. This is, of course, well known. It may be added that if the axes XYZ are moveable, and dX, dY, dZ denote the displacements in the directions of these axes, the formulae are still true.

Suppose that Ω is a function of X, Y, Z; say $\Omega = \Omega(X, Y, Z)$, we have

$$d\Omega = \frac{d\Omega}{dX}\,dX + \frac{d\Omega}{dY}\,dY + \frac{d\Omega}{dZ}\,dZ,$$

which may be written

$$\begin{aligned} d\Omega = {} & \Big(\alpha\,\frac{d\Omega}{dX} + \alpha'\,\frac{d\Omega}{dY} + \alpha''\,\frac{d\Omega}{dZ}\Big)\big(dX_{\prime} + \alpha\,dY + \alpha''\,dZ\big) \\ & + \Big(\beta\,\frac{d\Omega}{dX} + \beta'\,\frac{d\Omega}{dY} + \beta''\,\frac{d\Omega}{dZ}\Big)\big(\beta\,dX + \beta'\,dY + \beta''\,dZ\big) \\ & + \Big(\gamma\,\frac{d\Omega}{dX} + \gamma'\,\frac{d\Omega}{dY} + \gamma''\,\frac{d\Omega}{dZ}\Big)\big(\gamma\,dX + \gamma'\,dY + \gamma''\,dZ\big), \end{aligned}$$

and the coefficients of transformation α, β, γ, &c. being independent of X, Y, Z and $X_{\prime}$, $Y_{\prime}$, $Z_{\prime}$, the first factors are $\dfrac{d\Omega}{dX_{\prime}}$, $\dfrac{d\Omega}{dY_{\prime}}$, $\dfrac{d\Omega}{dZ_{\prime}}$, and we have

$$\begin{aligned} d\Omega = {} & \frac{d\Omega}{dX_{\prime}}\big[dX_{\prime} + (-r\,Y_{\prime} + q\,Z_{\prime})\,dt\big] \\ & + \frac{d\Omega}{dY_{\prime}}\big[dY_{\prime} + (-p\,Z_{\prime} + r\,X_{\prime})\,dt\big] \\ & + \frac{d\Omega}{dZ_{\prime}}\big[dZ_{\prime} + (-q\,X_{\prime} + p\,Y_{\prime})\,dt\big], \end{aligned}$$

which is a theorem assumed in the memoir.

The expressions of P, Q, R in the memoir depend upon the composition of several rotations. If, in Lamé's very convenient notation, we write

	$X_{\prime}$	$Y_{\prime}$	$Z_{\prime}$
X	α	β	γ
Y	α'	β'	γ'
Z	α''	β''	γ''

[PDF p. 377; printed p. 355.] to denote the before-mentioned relation between (X, Y, Z) and $(X_\prime, Y_\prime, Z_\prime)$, then the tables

	$X_\prime$	$Y_\prime$	$Z_\prime$
X	α	β	γ
Y	α'	β'	γ'
Z	α''	β''	γ''

	$X_\prime$	$Y_\prime$	$Z_\prime$
X	A	B	C
Y	A'	B'	C'
Z	A''	B''	C''

lead to the table

	$X_\prime$	$Y_\prime$	$Z_\prime$
X	$(\alpha, \beta, \gamma \succ A, A', A'')$	$(\alpha, \beta, \gamma \succ B, B', B'')$	$(\alpha, \beta, \gamma \succ C, C', C'')$
Y	$(\alpha', \beta', \gamma' \succ A, A', A'')$	$(\alpha', \beta', \gamma' \succ B, B', B'')$	$(\alpha', \beta', \gamma' \succ C, C', C'')$
Z	$(\alpha'', \beta'', \gamma'' \succ A, A', A'')$	$(\alpha'', \beta'', \gamma'' \succ B, B', B'')$	$(\alpha'', \beta'', \gamma'' \succ C, C', C'')$

where for shortness $(\alpha, \beta, \gamma \succ A, A', A'')$, &c., stand for $\alpha A + \beta A' + \gamma A''$, &c. these are of course the ordinary formulae for the composition of transformations from one set of rectangular axes to another. We may say that the coefficients of transformation from (X, Y, Z) to $(X_\prime, Y_\prime, Z_\prime)$ are (α, β, γ). And in dealing with several sets of coordinates, the coefficients of transformation may be distinguished by suffixes. I assume that the coefficients of transformation

from (x_0, y_0, z_0) to (x_1, y_1, z_1) are $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{01}$,

$$(x_1, y_1, z_1) \text{ to } (x_2, y_2, z_2) \quad (\mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{12},$$

$$(x_2, y_2, z_2) \text{ to } (x_3, y_3, z_3) \quad (\mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{23},$$

where $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{01}$, $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{12}$, $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{23}$ are to be considered as above mentioned. The relations p, q, r may be distinguished by suffixes in like manner, viz., $(p, q, r)_{01}$ will denote the rotations from (x_0, y_0, z_0) to (x_1, y_1, z_1), &c.

There is no difficulty in obtaining that

$$\begin{aligned}
p_{02} &= p_{01} + (\mathfrak{a}, \mathfrak{b}, \mathfrak{c} \succ \mathfrak{a}', \mathfrak{b}', \mathfrak{c}')_{12}\,(p, q, r)_{12}, \\
q_{02} &= q_{01} + (\mathfrak{a}', \mathfrak{b}', \mathfrak{c}' \succ \mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{12}\,(p, q, r)_{12}, \\
r_{02} &= r_{01} + (p, q, r)_{01}\,(p, q, r)_{12},
\end{aligned}$$

[PDF p. 378; printed p. 356.] and then by a repetition of the like transformation

$$p_{03} = p_{02} + (\mathfrak{a}, \mathfrak{b}, \mathfrak{c} \succ \mathfrak{a}', \mathfrak{b}', \mathfrak{c}')_{23}\,(p, q, r)_{23},$$

$$q_{03} = q_{02} + (\mathfrak{a}', \mathfrak{b}', \mathfrak{c}' \succ \mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{23}\,(p, q, r)_{23},$$

$$r_{03} = r_{02} + (p, q, r)_{02}\,(p, q, r)_{23},$$

and in like manner for any number of systems, but for the present purpose this is enough, as p_{03}, q_{03}, r_{03} are in fact the P, Q, R of the memoir. And if θ', ϖ', ϕ', Θ, Σ, Φ, v, y signify as before, then the coefficients of transformation from (x_0, y_0, z_0) to (x_1, y_1, z_1) are given by the table

	x_1	y_1	z_1
x_0	$\cos\varpi'\cos\theta' + \sin\varpi'\sin\theta'\cos\phi'$	$\sin\varpi'\cos\theta' - \cos\varpi'\sin\theta'\cos\phi'$	$\sin\theta'\sin\phi'$
y_0	$\cos\varpi'\sin\theta' - \sin\varpi'\cos\theta'\cos\phi'$	$\sin\varpi'\sin\theta' + \cos\varpi'\cos\theta'\cos\phi'$	$-\cos\theta'\sin\phi'$
z_0	$-\sin\varpi'\sin\phi'$	$\cos\varpi'\sin\phi'$	$\cos\phi'$

and the corresponding rotation coefficients are

$$\begin{aligned} p_{01} &= -\sin\varpi'\sin\phi'\,d\theta' + \cos\varpi'\,d\phi', \\ q_{01} &= \cos\varpi'\sin\phi'\,d\theta' + \sin\varpi'\,d\phi', \\ r_{01} &= -d\varpi' + \cos\phi'\,d\theta'. \end{aligned}$$

The coefficients of transformation from (x_1, y_1, z_1) to (x_2, y_2, z_2) are given by the table

	x_2	y_2	z_2
x_1	$\cos\Sigma\cos\Theta + \sin\Sigma\sin\Theta\cos\Phi$	$\sin\Sigma\cos\Theta - \cos\Sigma\sin\Theta\cos\Phi$	$\sin\Theta\sin\Phi$
y_1	$\cos\Sigma\sin\Theta - \sin\Sigma\cos\Theta\cos\Phi$	$\sin\Sigma\sin\Theta + \cos\Sigma\cos\Theta\cos\Phi$	$-\cos\Theta\sin\Phi$
z_1	$-\sin\Sigma\sin\Phi$	$\cos\Sigma\sin\Phi$	$\cos\Phi$

and the corresponding rotation coefficients are

$$\begin{aligned} p_{12} &= -\sin\Sigma\sin\Phi\,d\Theta + \cos\Sigma\,d\Phi, \\ q_{12} &= \cos\Sigma\sin\Phi\,d\Theta + \sin\Sigma\,d\Phi, \\ r_{12} &= -d\Sigma + \cos\Phi\,d\Theta. \end{aligned}$$

[PDF p. 379; printed p. 357.] and the coefficients of transformation from (x_2, y_2, z_2) to (x_3, y_3, z_3) are given by the table

	x_3	y_3	z_3
x_2	$\cos v \cos y$	$-\sin v$	$-\cos v \sin y$
y_2	$\sin v \cos y$	$\cos v$	$-\sin v \sin y$
z_2	$\sin y$	0	$\cos y$

and the corresponding rotation coefficients are given by

$$\begin{aligned} p_{23} &= 0, \\ q_{23} &= -dy, \\ r_{23} &= dv \cdot \sin y. \end{aligned}$$

From the last two tables it is easy to deduce the coefficients of transformation from (x_1, y_1, z_1) to (x_3, y_3, z_3), these are given by the table

	x_3	y_3	z_3
x_1	$\cos y\big[\cos(v-\varSigma)\cos\Theta - \sin(v-\varSigma)\sin\Theta\cos\varPhi\big] + \sin y \sin\Theta\sin\varPhi$	$-\sin(v-\varSigma)\cos\Theta - \cos(v-\varSigma)\sin\Theta\cos\varPhi$	$\cdots$
y_1	$\cos y\big[\cos(v-\varSigma)\sin\Theta + \sin(v-\varSigma)\cos\Theta\cos\varPhi\big] - \sin y \cos\Theta\sin\varPhi$	$-\sin(v-\varSigma)\sin\Theta + \cos(v-\varSigma)\cos\Theta\cos\varPhi$	$\cdots$
z_1	$\cos y \sin(v-\varSigma)\sin\varPhi + \sin y\cos\varPhi$	$\cos(v-\varSigma)\sin\varPhi$	$\cdots$

It is not easy to form the values of P, Q, R. We have

$$\begin{aligned} P\,dt = \sin y\,dv \\ &+ \big\{\cos y\big[\cos(v-\varSigma)\cos\Theta - \sin(v-\varSigma)\sin\Theta\cos\varPhi\big] + \sin y\sin\Theta\sin\varPhi\big\}\big(-\sin\varSigma\sin\varPhi\,d\Theta + \cos\varSigma \\ &+ \big\{\cos y\big[\cos(v-\varSigma)\sin\Theta + \sin(v-\varSigma)\cos\Theta\cos\varPhi\big] - \sin y\cos\Theta\sin\varPhi\big\}\big(\cos\varSigma\sin\varPhi\,d\Theta + \sin\varSigma\,d \\ &+ \big\{\cos y\sin(v-\varSigma)\sin\varPhi + \sin y\cos\varPhi\big\}\big(-d\varSigma + \cos\varPhi\,d\Theta\big)\ldots \end{aligned}$$

[PDF p. 380; printed p. 358.]

$$
\begin{aligned}
&\quad + \cos v \cos y\,(-\sin \varpi' \sin \phi'\, d\theta' + \cos \varpi'\, d\phi') \\
&\quad + \sin v \cos y\,(\cos \varpi' \sin \phi'\, d\theta' + \sin \varpi'\, d\phi') \\
&\quad + \sin y\,(-d\varpi' + \cos \phi'\, d\theta'),
\end{aligned}
$$

$$
\begin{aligned}
Q\,dt &= -dy \\
&\quad + \big[-\sin(v-\varSigma)\cos\Theta - \cos(v-\varSigma)\sin\Theta\cos\varPhi\big]\big(-\sin\varSigma\sin\varPhi\,d\Theta + \cos\varSigma\,d\varPhi\big) \\
&\quad + \big[-\sin(v-\varSigma)\sin\Theta + \cos(v-\varSigma)\cos\Theta\cos\varPhi\big]\big(\cos\varSigma\sin\varPhi\,d\Theta + \sin\varSigma\,d\varPhi\big) \\
&\quad + \big[\cos(v-\varSigma)\sin\varPhi\big]\big(-d\varSigma + \cos\varPhi\,d\Theta\big) \\
&\quad - \sin v\,(-\sin \varpi' \sin \phi'\, d\theta' + \cos \varpi'\, d\phi') \\
&\quad + \cos v\,(\cos \varpi' \sin \phi'\, d\theta' + \sin \varpi'\, d\phi'),
\end{aligned}
$$

$$
\begin{aligned}
R\,dt &= \cos y\,dv \\
&\quad + \big[-\sin(v-\varSigma)\cos\Theta\cos\varPhi + \cos y\sin\Theta\sin\varPhi\big]\,d\Theta + \ldots \\
&\quad + \big[\sin y\sin(v-\varSigma)\sin\varPhi\big]\,d\varPhi \\
&\quad + \cos y\big[-d\varSigma + \cos\varPhi\,d\Theta\big] \\
&\quad - \cos v \sin y\,(-\sin \varpi' \sin \phi'\, d\theta' + \cos \varpi'\, d\phi') \\
&\quad - \sin v \sin y\,(\cos \varpi' \sin \phi'\, d\theta' + \sin \varpi'\, d\phi') \\
&\quad + \cos y\,(-d\varpi' + \cos \phi'\, d\theta'),
\end{aligned}
$$

which are at once reducible into the before-mentioned form.

It only remains to prove the expressions for $\dfrac{d\Omega}{dx_3}$, $\dfrac{d\Omega}{dy_3}$, $\dfrac{d\Omega}{dz_3}$ assumed in the memoir. The relations between (x_2, y_2, z_2) and (x_3, y_3, z_3) give

$$
\begin{aligned}
\frac{d\Omega}{dx_3} &= \cos v\cos y\,\frac{d\Omega}{dx_2} + \sin v\cos y\,\frac{d\Omega}{dy_2} + \sin y\,\frac{d\Omega}{dz_2}, \\
\frac{d\Omega}{dy_3} &= -\sin v\,\frac{d\Omega}{dx_2} + \cos v\,\frac{d\Omega}{dy_2}, \\
\frac{d\Omega}{dz_3} &= -\cos v\sin y\,\frac{d\Omega}{dx_2} - \sin v\sin y\,\frac{d\Omega}{dy_2} + \cos y\,\frac{d\Omega}{dz_2};
\end{aligned}
$$

these formulae apply to any point whatever the coordinates of which to the one set of axes are (x_2, y_2, z_2) and to the other set (x_3, y_3, z_3) and in obtaining them the coefficients of transformation v, y are of course considered as independent of either set of coordinates. But the coordinates (x_2, y_2, z_2) of the body are

$$x_2 = r\cos y\cos v, \qquad y_2 = r\cos y\sin v, \qquad z_2 = r\sin y;$$

[PDF p. 381; printed p. 359.] and using the foregoing equations in order to introduce into the formulae the differential coefficients $\frac{d\Omega}{dr}, \frac{d\Omega}{dv}, \frac{d\Omega}{dy}$ in the place of $\frac{d\Omega}{dx_2}, \frac{d\Omega}{dy_2}, \frac{d\Omega}{dz_2}$, we find

$$\begin{aligned}
\frac{d\Omega}{dx_2} &= \cos v \cos y \,\frac{d\Omega}{dr} - \frac{1}{r}\sin v \sec y \,\frac{d\Omega}{dv} - \frac{1}{r}\cos v \sin y \,\frac{d\Omega}{dy},\\
\frac{d\Omega}{dy_2} &= \sin v \cos y \,\frac{d\Omega}{dr} + \frac{1}{r}\cos v \sec y \,\frac{d\Omega}{dv} - \frac{1}{r}\sin v \sin y \,\frac{d\Omega}{dy},\\
\frac{d\Omega}{dz_2} &= \sin y \,\frac{d\Omega}{dr} + \frac{1}{r}\cos y \,\frac{d\Omega}{dy}.
\end{aligned}$$

Substituting these values we have the required formulae

$$\frac{d\Omega}{dx_3} = \frac{d\Omega}{dr}, \qquad \frac{d\Omega}{dy_3} = \frac{1}{r}\sec y \,\frac{d\Omega}{dv}, \qquad \frac{d\Omega}{dz_3} = \frac{1}{r}\frac{d\Omega}{dy},$$

where on the right-hand side $\Omega = \Omega(r, v, y, \Theta, \mathit{\Sigma}, \mathit{\Phi}, \theta', \varpi', \phi')$.

2, Stone Buildings, W.C. 28th Dec., 1858.

216. Tables of the Developments of Functions in the Theory of Elliptic Motion

[PDF p. 382; printed p. 360.]

[From the *Memoirs of the Royal Astronomical Society*, vol. XXIX. (1861), pp. 191–306. Read January 12, 1859.]

The notation made use of is as follows; viz., r, f, a, e, g denote

r, the radius vector;
f, the true anomaly;
a, the mean distance;
e, the eccentricity;
g, the mean anomaly;

so that

$$\frac{r}{a} = \mathrm{elqr}(e, g),$$

and

$$f = \mathrm{elta}(e, g),$$

are known functions of e, g. Moreover, x denotes the periodic part of the quotient radius $\frac{r}{a}$, and y the equation of the centre, or periodic part of the true anomaly f; so that

$$\frac{r}{a} = 1 + x,$$

$$f = g + y,$$

and (x, y) are therefore also known functions of e, g.

[PDF p. 383; printed p. 361.] Formulae for the development in multiple cosines or sines, up to the terms in e^7, of the functions

$$\left(x^1, x^2, \ldots x^7\right),$$

$$\left(x^a, x^{a+1}, \ldots x^{a+7}\right)\begin{matrix}\cos\\\sin\end{matrix} jf, \quad j = 1 \text{ to } 7,$$

$$\left(\left(\frac{r}{a}\right)^{+1}, \left(\frac{r}{a}\right)^{-1}, \ldots \left(\frac{r}{a}\right)^{+5}, \left(\frac{r}{a}\right)^{-5}\right), \quad \log\frac{r}{a}, \ldots$$

$$\left(\left(\frac{r}{a}\right)^{+1}, \left(\frac{r}{a}\right)^{-1}, \ldots \left(\frac{r}{a}\right)^{+5}, \left(\frac{r}{a}\right)^{-5}\right)\begin{matrix}\cos\\\sin\end{matrix} jf, \quad j = 1 \text{ to } 5,$$

where j is an indeterminate symbol, are given by Leverrier in the *Annales de l'Observatoire de Paris*, tom. I. (1855), pp. 346–348. It occurred to me that it would be desirable to form tables such as are contained in the present memoir, of various functions of the forms

$$x^\alpha \begin{matrix}\cos\\\sin\end{matrix} jf, \quad \left(\frac{r}{a}\right)^\alpha \begin{matrix}\cos\\\sin\end{matrix} jf,$$

obtainable from Leverrier's formulae by assigning particular integer values to the symbol j. The calculations were performed under my superintendence by Messrs Greedy and Davis; and I have to acknowledge my obligation for a grant to defray the expenses, made to me out of the Donation Fund at the disposal of the Council of the Royal Society.

A cosine series is in general represented in the form

$$\sum [\cos]^i \cos ig,$$

where i extends from $-\infty$ to $+\infty$, and the coefficients $[\cos]^i$ satisfy the condition

$$[\cos]^{-i} = [\cos]^i;$$

and in like manner a sine series is represented in the form

$$\sum [\sin]^i \sin ig,$$

where i extends from $-\infty$ to $+\infty$, and the coefficients $[\sin]^i$ satisfy the condition

$$[\sin]^{-i} = -[\sin]^i, \quad \text{and in particular} \quad [\sin]^0 = 0.$$

Thus $[\cos]^i$ is a general symbol denoting the coefficient of $\cos ig$, in a cosine series considered as involving as well negative as positive multiples of the argument g, or, what is the same thing, $[\cos]^0$ is the constant term and $[\cos]^i$ is the half coefficient of $\cos ig$, when the series is considered as containing only positive multiples of the argument. And in like manner $[\sin]^i$ is a general symbol for the coefficient of $\sin ig$, in a sine series considered as involving as well negative as positive multiples of the argument g, or, what is the same thing, $[\sin]^i$ is the half coefficient of $\sin ig$, when the series is considered as containing only positive multiples of the argument.

[PDF p. 384; printed p. 362.] In the case of a pair of corresponding functions $x^m \cos jf$ and $x^m \sin jf$, or $\left(\frac{r}{a}\right)^m \cos jf$ and $\left(\frac{r}{a}\right)^m \sin jf$, one of them expanded in the form

$$\sum [\cos]^i \cos ig,$$

and the other in the form

$$\sum [\sin]^i \sin ig,$$

the sums and differences of the corresponding coefficients $[\cos]^i$, $[\sin]^i$, are represented by the notation

$$[\cos \pm \sin]^i,$$

and it is clear that we have

$$[\cos + \sin]^{-i} = [\cos - \sin]^i, \qquad [\cos \pm \sin]^0 = [\cos]^0.$$

These coefficients $[\cos \pm \sin]^i$ are for many purposes equally useful with the coefficients $[\cos]^i$, $[\sin]^i$, and they are here tabulated accordingly.

The functions tabulated are as follows:

$$\begin{aligned}
&\left(x^1, x^2, \ldots x^7\right) \quad , \quad [\cos], \\
&\left(x^a, x^{a+1}, \ldots x^{a+7}\right)_{\sin}^{\cos} jf,\ j = 1 \text{ to } 7, \quad [\cos], [\sin], [\cos \pm \sin], \\
&\left(\left(\frac{r}{a}\right)^{+1}, \left(\frac{r}{a}\right)^{-1}, \ldots \left(\frac{r}{a}\right)^{+5}, \left(\frac{r}{a}\right)^{-5}\right),\ \log\frac{r}{a}, \ldots, \quad [\cos], \\
&\left(\left(\frac{r}{a}\right)^{+1}, \left(\frac{r}{a}\right)^{-1}, \ldots \left(\frac{r}{a}\right)^{+5}, \left(\frac{r}{a}\right)^{-5}\right)_{\sin}^{\cos} jf,\ j = 1 \text{ to } 5, \quad [\cos], [\sin], [\cos \pm \sin],
\end{aligned}$$

all the developments being carried up to e^7, the limit of the formulas from which they are deduced.

The calculations are based upon Leverrier's formulae for

$$\left(x^a, x^{a+1}, \ldots x^{a+7}\right)_{\sin}^{\cos} jf,$$

which are made use of under the following forms, which I call the Datum Tables

$$\left(x^a, x^{a+1}, \ldots x^{a+7}\right)_{\sin}^{\cos} jf.$$

[PDF p. 385; printed p. 363.]

Datum Tables of $(x^a, \dots x^{a+7})\cos jf$ *and* $\cos jf$ *extracts.*

Cosine Table $\frac{1}{a}$ of	$\left[\frac{1}{a}\right]$ $\frac{e^a}{\lvert 0}$	$\frac{e^{a+1}}{\lvert 2 \lvert 0}$	$\frac{e^{a+2}}{\lvert 4 \lvert 0}$	$\frac{e^{a+3}}{\lvert 6 \lvert 0}$	$\left[\frac{2}{a}\right]$ $\frac{e^a}{\lvert 0}$	$\frac{e^{a+1}}{\lvert 2 \lvert 0}$	$\frac{e^{a+2}}{\lvert 4 \lvert 0}$	
$x^0 \cos jf$	$+1$	$-8j^2$	$+28j^4 - 1280j^2$	$\dots$	0	$-3j^2$	$+89j^2 - 110j^4$	$+18$
$x^1 \cos jf$		$-2j^2$	$-3j^2 + 71j^4$	$\dots$		-1		
$x^2 \cos jf$	$+2$	$-2j^2$	$+1900j^2 - \dots$		-1	$+13j^2$	$-263j^4 - \dots$	
$x^3 \cos jf$		$+2$	$-19j^2 + 1090j^2$		-9	$+579j^2 - \dots$	$-1125j^4 + \dots$	
$x^4 \cos jf$		$+4$	$-49j^2 + 1502j^2 - 1320j^4$	$\dots$		$+50j^2$		
$x^5 \cos jf$			$+144$	$-1980j^2$			$+480$	
$x^6 \cos jf$				$+441$				
$x^7 \cos jf$				$+1400$				
$x^8 \cos jf$				0				
Datum Table $(x^1, \dots x^8)\cos jf \dots$								

[*Numerical entries reproduce the printed Datum Tables; columns correspond to coefficients of successive powers of* e^a, e^{a+1}, *etc. in the cosine expansion of* $x^n \cos jf$.]

[PDF p. 386; printed p. 364.]

Datum Table $(x^a, \dots x^{a+7}) \cos jf$, *continued.*

Cosine Table	$\left[\frac{1}{a}\right]$				$\left[\frac{2}{a}\right]$		
$\frac{1}{a}$ of	$\frac{e^a}{\lvert 2 \rvert 0}$	$\frac{e^{a+1}}{\lvert 4 \rvert 0}$	$\frac{e^{a+2}}{\lvert 6 \rvert 0}$	$\dots$	$\frac{e^a}{\lvert 2 \rvert 0}$	$\frac{e^{a+1}}{\lvert 4 \rvert 0}$	$\frac{e^{a+2}}{\lvert 6 \rvert 0}$
$x^0 \cos jf$	$+4j^2$	$-64j^2 + 296j^4$	$-2104j^2$	$\dots$	$+3j^2$	$-1100j^2 + \dots$	$+30163j^2 + \dots$
$x^1 \cos jf$		$+64j^2 - 1981j^2$	$-12247j^2 + \dots$	$\dots$		$+1742j^2 - \dots$	$\dots$
$x^2 \cos jf$	$+1$	$-2j^2$	$+440j^2$			$+613$	$-90209j^2$
$x^3 \cos jf$		$+5$	$-67j^2 + 1908j^2$		-12	$+845j^2 - \dots$	$+45980j^2 + \dots$
$x^4 \cos jf$			$+4$	$-1880j^2$			$+480j^2 - \dots$
$x^5 \cos jf$				-6	-12		$+540$
$x^6 \cos jf$				-96			$+540$
$x^7 \cos jf$				-3500			
$x^8 \cos jf$				0			
Datum Table $(x^1, \dots x^8) \cos jf \dots$							

[PDF p. 387; printed p. 365.]

Datum Table $(x^a, \ldots x^{a+7}) \cos jf$, *concluded.*

Cosine Table $\frac{1}{a}$ of	$\left[\frac{1}{a}\right]$ $\frac{e^a}{\lfloor 4 \rfloor 0}$	$\frac{e^{a+1}}{\lfloor 6 \rfloor 0}$	$\left[\frac{2}{a}\right]$ $\frac{e^a}{\lfloor 4 \rfloor 0}$	$\frac{e^{a+1}}{\lfloor 6 \rfloor 0}$	$\left[\frac{3}{a}\right]$ $\frac{e^a}{\lfloor 4 \rfloor 0}$	$\frac{e^{a+1}}{\lfloor 6 \rfloor 0}$	
$x^0 \cos jf$	$+16j^2$	$-384j^2$	$+4096^a - 1180j^4$	$-50450j^2 - \ldots$	$+643j^4 + \ldots$	$+1883j^4 - \ldots$	$-$
$x^1 \cos jf$		-84	$-3137 \ldots$	$-13125j^2 - \ldots$	$-1273j^2 + \ldots$	$+10193j^2 + \ldots$	$+$
$x^2 \cos jf$	$+48$	$-7008j^2$	$+1480j^2$	$-3360j^2 + \ldots$	$-127j^2 + \ldots$	$-1100j^2 + \ldots$	$+1$
$x^3 \cos jf$		$+96$	$-15080j^2 + \ldots$	$+1450j^2 + \ldots$	$+540j^2 + \ldots$	$-1422j^2 + \ldots$	$+$
$x^4 \cos jf$			$+75$	$-1242j^2 + \ldots$	$-540j^2 + \ldots$	$+1450j^2 + \ldots$	$-$
$x^5 \cos jf$				$+ \ldots$	-100	$-1320j^2 + \ldots$	$-$
$x^6 \cos jf$					-7000	$+3800$	
$x^7 \cos jf$					-4200	$+700$	
$x^8 \cos jf$							
Datum Table $(x^1, \ldots x^8) \cos jf \ldots$							

[End of pp. 351–387 chunk; paper 216 continues in next chunk (tables of developments to e^7).]